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Small Signal Stability of Grid-Forming Control for Doubly-fed Induction Generators: A comparative study of different control algorithms

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ABSTRACT This paper presents a comprehensive small-signal stability analysis of three widely used grid-forming (GFM) control strategies, namely droop control, virtual synchronous generator (VSG), and generalized VSG (GVSG), implemented on a doubly-fed induction generator (DFIG) based wind turbine. Modal analysis is performed to uncover critical oscillatory dynamics, in terms of frequency and damping characteristics, under various realistic operating conditions, considering the impacts of grid strength, virtual impedance, inner-current control loops, output power levels, and off-grid operation. To validate and complement the modal analysis findings, time-domain simulations under small disturbances are performed. Based on the results from combined frequency and time-domain analysis, the study further compares the respective strengths and limitations of each control approach, providing valuable insights for the integration of renewable energy systems in future power grids.

INDEX TERMS DFIG, droop control, generalized virtual synchronous generator, grid-following (GFL) control, grid-forming control, small-signal stability, virtual synchronous generator, wind energy

I. INTRODUCTION

A significant transformation in power systems mainly has been driven by the worldwide efforts to reduce greenhouse gas emissions, with power energy sectors increasingly transitioning toward renewable energy sources [1]. This involves a gradual displacement of conventional synchronous generators by renewable energy technologies with many grids such as Germany and Denmark experiencing a large portion of the generation mix from these clean sources [2]. Among these, wind turbines and photovoltaic (PV) have emerged as the most dominant due to their increasingly mature and affordable technologies [1].

A critical component for the renewable energy integration is the power inverters, serving as the interface between renewable energy systems and the grid. Currently, the inverters used for renewable energy integration are predominantly configured in grid-following (GFL) mode, relying on a phase-locked loop (PLL) to track grid voltages for

synchronization [3]. The active and reactive power injections are determined by the estimated grid voltage and power references typically obtained from higher-level controllers such as a maximum power point tracking (MPPT) algorithm [4], [5]. With this concept, the inverter does not regulate its own voltage and frequency in the similar way conventional synchronous generators normally operate, making it act similar to a current source which limits grid support [6]. Further problems include heavy reliance on PLL, which may lose synchronization under weak-grid [7] or unbalanced condition [8], and lack of inertial support during frequency events, leading to a large frequency drop in high level renewable energy penetration [9].

To address these challenges, GFM control is adopted, enabling the inverter to function as a voltage source that actively generates its own voltage and frequency. One common implementation is to emulate the behavior of a synchronous generator [10], although other grid-forming strategies also

exist. Hence, this concept enables the inverter to facilitate inertia provision [10], stability enhancement under weak-grids [11], black-start capability [12], and islanding capability [13].

Currently, GFM inverters have been realized using a variety of control architectures. One of the earliest GFM implementations is droop control [14]–[16], where the inverter's voltage and frequency are generated in proportion to real and reactive power deviations from pre-specified references. Thus, the droop aims to emulate power-sharing characteristics of synchronous generators. Although widely adopted for its simplicity, the droop is known for its lack of inertia support [17]. To overcome this problem, it is necessary to explicitly mimic the swing equation of synchronous generators. For example, the authors in [18] proposed the concept of VSG, while synchronverter has been introduced in [19]. Despite its capability of providing inertia, tuning of VSG parameters remains challenging. Specifically, the performance of VSG is determined by two somewhat conflicting parameters, namely inertia and droop coefficient [20]. Incorrect selection of these constants could lead to poor transient responses [21]. To address this issue, a GVSG introduces a lead–lag compensator in series with the inertia constant, enabling more flexible tuning of both constants [9].

Significant efforts have also been made to implement GFM algorithms for specific renewable energy resources. In [22] and [23], an energy storage system (ESS) with VSG control has been added to a PV system. In [24], advanced current limiting and power-sharing controls are added in a PV-based GFM. Refs. [25] and [26] have implemented the droop control and virtual synchronous generator algorithms, respectively, on a DFIG. Further, GFM algorithms have also been implemented in a permanent magnet synchronous generator (PMSG) [27], [28].

Despite their potential features to enhance grid stability, GFM inverters remain in the early stage of development, with no implementation standards. Therefore, their stability assessment is of paramount importance. In general, the stability assessment methods can be broadly categorized into time- and frequency-domain approaches. In the former, the dynamic response of the inverter system is studied under disturbances to identify potential instabilities. The time-domain responses can be obtained from software simulations [29], hardware-in-the-loop (HIL) simulations [30], or hardware realization [31]. In the latter, the system stability is investigated by analyzing frequency-domain indices. For instance, the impedance-based method assesses stability by analyzing the interaction between inverter and grid impedances [6]. In [32], [33], this method has been used to investigate the sub-synchronous resonance (SSR) of a GFM DFIG. Another key frequency-domain method is small-signal stability analysis, which linearizes the system around a steady-state operating condition to reveal modal properties, enabling a detailed assessment of oscillatory behavior. In [34], modal analysis is employed to identify slow interaction modes between GFL and GFM in a distribution system, while Ref. [35] conducts

eigenvalue analysis on virtual synchronous machine implementations on DFIGs.

Despite the above studies, important research gaps still exist. Only a few works have compared different GFM control methods implemented on DFIGs, even though DFIGs are the most widely used type of wind turbines. Among these limited studies, the focus has typically been on comparative analyses of transient responses [26], [35] or impedance characteristics [17]. In contrast, small-signal stability analysis has been minimally explored for stability analysis of GFM DFIGs despite its effectiveness in revealing detailed oscillatory behavior. Building upon these gaps, this paper aims to perform a comparative small-signal stability analysis on three widely used GFM algorithms, namely droop, VSG, and GVSG controls, implemented on DFIGs. To the best knowledge of the authors, this is the first study that systematically compares droop, VSG, and GVSG control strategies for DFIGs within a unified small-signal stability framework. Unlike prior works limited to specific conditions, the proposed study evaluates stability across a wide range of realistic operating scenarios under both frequency- and time-domain perspectives. Specifically, the contributions of this paper are:

- 1) To implement three commonly used algorithms, namely droop, VSG, and GVSG control strategies, for DFIG control. While droop and VSG have previously been implemented in DFIGs, their effectiveness from a small-signal stability perspective has not been fully explored. Thus, this work extends prior studies by not only re-examining droop and VSG in this context but also providing the first systematic evaluation of GVSG effectiveness in DFIG operation.
- 2) While implementing the three GFM control strategies, a modified approach, inspired from [36], is adopted to ensure minimal hardware changes to the conventional GFL DFIG control structure. Unlike most existing approaches in the literature [26], [33], [37], which often employ entirely different control architectures, the proposed method leverages the similarity with the well-established GFL framework. This makes the transition from GFL to GFM operation more practical and cost-effective, since the majority of deployed DFIGs already operate in GFL mode and can be retrofitted with minimal hardware modifications, thereby reducing capital investment.
- 3) To perform a comprehensive small-signal analysis in the frequency-domain to reveal oscillatory behavior of three GFM control algorithms implemented on DFIG. The analysis identifies the key oscillatory modes and systematically investigates the merits and demerits of each control strategy under various realistic operating scenarios, offering valuable insights into their stability characteristics.
- 4) Lastly, to validate and complement the findings of the frequency-domain analysis, time-domain simulations

under small disturbances are performed. These simulations provide further insight into the system's dynamic responses.

The remainder of the paper is organized as follows. Sections II and III respectively present the DFIG model under GFL and GFM control strategies. The network model is described in Section IV. A self-contained discussion on modal analysis is provided in Section V, followed by the test system description in Section VI. The analysis of the GFM-controlled DFIG using modal analysis and time-domain simulation is presented in Sections VII and VIII, respectively. Finally, key findings are summarized in Section IX.

II. DFIG MODELLING WITH GRID FOLLOWING CONTROL

In this section, the dynamic models of different DFIG components are described. The model is modified from [38] and encompasses the wind turbine, induction machine, RSC, GSC, drive train, terminal voltage dynamics, PLL dynamics, DC-link dynamics, and the GSC filter dynamics as can be seen from Fig. 1. In this configuration, the DFIG operates in GFL mode due to the fact that the system is synchronized using PLL.

A. TURBINE MODEL

A wind turbine captures kinetic energy from the wind. This kinetic energy is converted into mechanical power through the rotation of the turbine blades. The mechanical power is then used to drive an electrical generator, producing electricity. In reality, this energy conversion involves complex aerodynamic processes that can be accurately described by partial differential equations based on fluid dynamics. However, such detailed modeling is not required for power system stability studies as it is computationally demanding. Instead, the turbine can be effectively represented using an algebraic equations. Suppose the wind flows at speed v_w , then the turbine torque T_m can be expressed as follows:

$$T_m = C_p(\lambda_{\text{TSR}}, \beta) 0.5 \rho \pi R_b^2 v_w^3 / \omega_t \quad (1)$$

The turbine's rotational speed is represented by ω_t . The constants ρ and R_b refer to the air density and the blade length, respectively. The turbine performance coefficient $C_p(\lambda_{\text{TSR}}, \beta)$ depends on the tip speed ratio $\lambda_{\text{TSR}} = \omega_t R_b / v_w$ and the pitch angle β .

B. INDUCTION GENERATOR MODEL

The induction generator is modeled as a fourth-order dynamic system by assuming that the DFIG operates under balanced conditions [39], resulting in the following state-space representation:

$$\dot{x}_{gen} = \mathbf{A}_{gen} x_{gen} + \mathbf{B}_{gen} u_{gen} \quad (2)$$

$$\mathbf{B}_{gen} = \begin{bmatrix} \frac{-X_s}{\omega_b} & 0 & \frac{X_m}{\omega_b} & 0 \\ 0 & \frac{-X_s}{\omega_b} & 0 & \frac{X_m}{\omega_b} \\ -\frac{X_m}{\omega_b} & 0 & \frac{X_r}{\omega_b} & 0 \\ 0 & \frac{-X_m}{\omega_b} & 0 & \frac{X_r}{\omega_b} \end{bmatrix}^{-1}$$

$$\mathbf{A}_{gen} = -\mathbf{B}_{gen} \begin{bmatrix} -r_s & -\omega_s X_s & 0 & \omega_s X_m \\ \omega_s X_s & -r_s & -\omega_s X_m & 0 \\ 0 & -s X_m & r_r & s X_r \\ s X_m & 0 & -s X_r & r_r \end{bmatrix}$$

where $s = 1 - \omega_r$ is the slip. ω_s and $\omega_b = 2\pi f$ denote the synchronous speed in per-unit and base speed, respectively. The parameters r_s and r_r refer to the stator and rotor resistances. The reactances X_s , X_r , and X_m represent the stator, rotor, and mutual reactances, respectively. The state vector is given by $x_{gen} = [i_{qs} \ i_{ds} \ i_{qr} \ i_{dr}]^T$, which includes the stator and rotor currents in the corresponding q - and d -axis. Finally, $u_{gen} = [v_{qs} \ v_{ds} \ v_{qr} \ v_{dr}]^T$ indicates the stator and rotor voltages in the corresponding q - and d -axis.

C. DRIVE-TRAIN MODEL

In a typical DFIG drive-train modeling, the blades and generator are connected through a flexible shaft or gearbox. Consequently, the two-mass model is widely used to accurately capture the drive-train dynamics. This model is represented by (3) as follows:

$$\dot{x}_{drive} = \mathbf{A}_{drive} x_{drive} + \mathbf{B}_{drive} u_{drive} \quad (3)$$

$$\mathbf{A}_{drive} = \begin{bmatrix} \frac{D_t - D_{Tg}}{2H_t} & \frac{D_{Tg}}{2H_t} & \frac{-1}{2H_t} \\ \frac{D_{Tg}}{2H_g} & \frac{D_t - D_{Tg}}{2H_g} & \frac{1}{2H_g} \\ K_{Tg}\omega_s & -K_{Tg}\omega_s & 0 \end{bmatrix}$$

$$\mathbf{B}_{drive} = \begin{bmatrix} \frac{1}{2H_t} & 0 & 0 \\ 0 & \frac{-1}{2H_g} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

where $x_{drive} = [\omega_t \ \omega_r \ T_g]^T$ is the state vector of the drive train and $u_{drive} = [T_m \ T_e \ 0]^T$ denotes the input vector to the drive-train. T_m , T_e , and T_g represent the mechanical torque acting on the turbine blades, the electrical torque in the generator, and the internal torque of the drive-train model, respectively. The turbine and generator inertia constants are given by H_t and H_g , respectively. The damping coefficients for the turbine, generator, and shaft coupling are represented by D_t , D_g , and D_{Tg} , respectively. Finally, K_{Tg} denotes the shaft stiffness coefficient.

D. GSC FILTER

The GSC filter is typically located between the inverter and the DFIG terminal, acting as a low-pass filter that filters out high-frequency harmonics generated by the pulse width modulation (PWM) switching operation. The dynamic equation of this filter, modeled using an RL component, is given as follows:

$$\dot{x}_{fil} = \mathbf{A}_{fil} x_{fil} + \mathbf{B}_{fil} u_{fil} \quad (4)$$

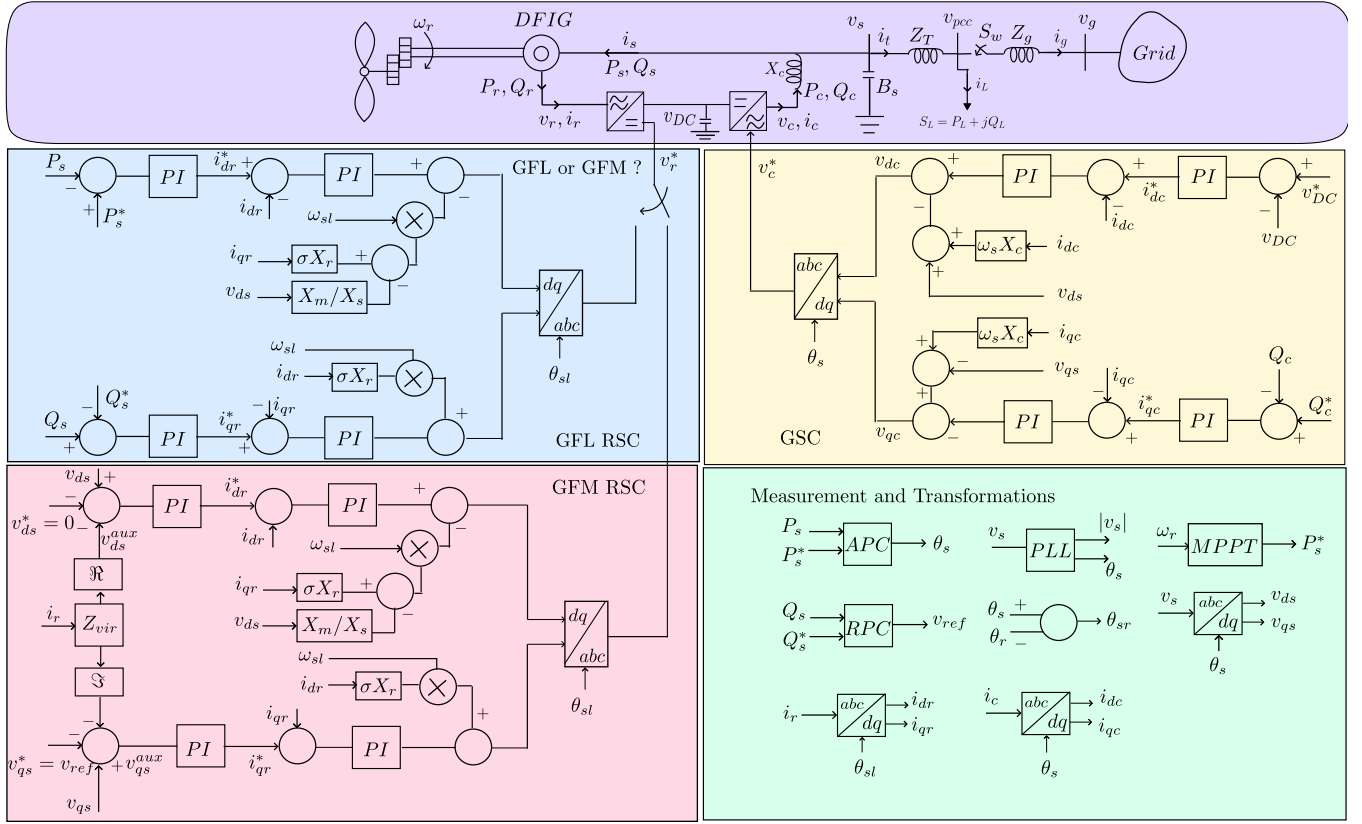


FIGURE 1. DFIG control configuration in GFL and GFM modes.

$$\mathbf{B}_{fil} = \begin{bmatrix} \frac{X_c}{\omega_b} & 0 \\ 0 & \frac{X_c}{\omega_b} \end{bmatrix}^{-1}$$

$$\mathbf{A}_{fil} = \begin{bmatrix} \frac{X_c}{\omega_b} & 0 \\ 0 & \frac{X_c}{\omega_b} \end{bmatrix}^{-1} \begin{bmatrix} 0 & \omega_s X_c \\ -\omega_s X_c & 0 \end{bmatrix}$$

where $x_{fil} = [i_{qc} \ i_{dc}]$ and $u_{fil} = [v_{qg} - v_{qs} \ v_{dg} - v_{ds}]$ represent the state and input vectors of GSC filter, respectively. X_g indicates the filter reactance. The states of the filter are i_{dc} and i_{qc} , denoting the filter current i_c in the d - and q -axis, respectively.

E. DC-LINK MODEL

The RSC and GSC are interconnected through the DC-link, which acts as a temporary energy storage system. The DC-link voltage is maintained constant by the GSC through the control of power exchange via the converter. The dynamics of the DC-link capacitor voltage are inherently nonlinear and depend on the instantaneous power balance between the RSC and GSC. These dynamics can be mathematically expressed as follows:

$$\dot{v}_{DC} = \frac{1}{C_{DC}v_{DC}}(P_r - P_c) \quad (5)$$

where the parameters C_{DC} and v_{DC} represent the DC-link capacitor and its corresponding voltage, respectively. Additionally, $P_r = v_{qr}i_{qr} + v_{dr}i_{dr}$ and $P_c = v_{qc}i_{qc} + v_{dc}i_{dc}$

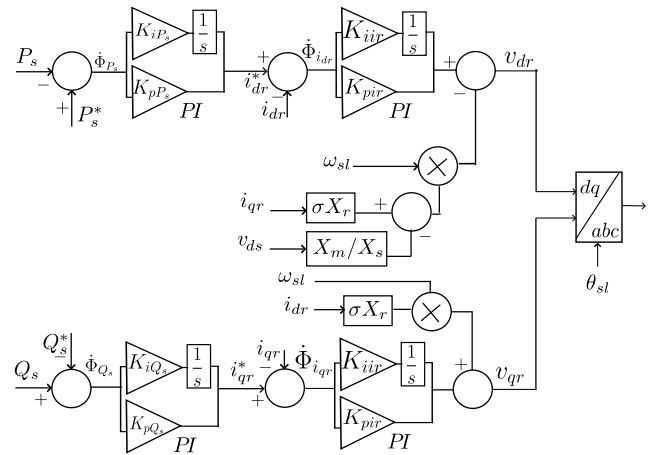


FIGURE 2. Rotor-side converter controller.

denote the power flowing into the DC-link from the RSC and the power flowing out of the DC-link through the GSC, respectively.

F. ROTOR-SIDE CONVERTER

The RSC is a critical component in a DFIG system, responsible for controlling the rotor voltage (see Fig. 2). To achieve this, the RSC controller uses a two-level cascaded structure comprising inner and outer control loops. The controller

operates in the d - q synchronous reference frame to enable independent control of the stator active power P_s and reactive power Q_s . The inner loop regulates the rotor currents to prevent overcurrent, while the outer loop generates reference signals for the inner loop. Specifically, the outer loop of the RSC controls the stator real power P_s , which is typically governed by a MPPT algorithm. In this paper, the model for the MPPT tracking is presented as follows:

$$P_s^* = K_{opt}\omega_r^3 \quad (6)$$

where P_s^* is the reference power for P_s and K_{opt} is the optimal power coefficient expressed in per-unit. With this configuration as shown in Fig. 2, the dynamic model of RSC is as follows:

$$\begin{aligned} \dot{x}_{rsc} &= \mathbf{A}_{rsc}x_{rsc} + \mathbf{B}_{rsc}u_{rsc} \quad (7) \\ \mathbf{A}_{rsc} &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ K_{iP_s} & 0 & 0 & 0 \\ 0 & K_{iQ_s} & 0 & 0 \end{bmatrix} \\ \mathbf{B}_{rsc} &= \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ K_{pP_s} & -K_{pP_s} & 0 & 0 & -1 & 0 \\ 0 & 0 & K_{pQ_s} & -K_{pQ_s} & 0 & -1 \end{bmatrix} \end{aligned}$$

where $x_{rsc} = [\Phi_{P_s} \ \Phi_{Q_s} \ \Phi_{i_{qr}} \ \Phi_{i_{dr}}]^T$ and $u_{rsc} = [P_s^* \ P_s \ Q_s \ Q_s^* \ i_{qr} \ i_{dr}]^T$. The states $\Phi_{i_{dr}}$ and $\Phi_{i_{qr}}$ are associated to the inner-loop current controllers, whereas the states Φ_{P_s} and Φ_{Q_s} are related to the outer-loop controllers. The parameters K_{pP_s} , K_{iP_s} , K_{pQ_s} and K_{iQ_s} denote the relevant constants in the RSC PI controllers.

Finally, the rotor voltages algebraic equations v_{dr} and v_{qr} can be obtained from the RSC controller in Fig. 2. From the inner-loop current controllers, it can be obtained the following :

$$\begin{aligned} v_{dr} &= K_{pir}(i_{dr}^* - i_{dr}) + K_{iir}\Phi_{i_{dr}} \\ &\quad - \omega_{sl} \left(\sigma X_r i_{qr} - \frac{X_m}{X_s} v_{ds} \right) \quad (8) \\ v_{qr} &= K_{pir}(i_{qr}^* - i_{qr}) + K_{iir}\Phi_{i_{qr}} + \omega_{sl} \cdot \sigma X_r i_{dr} \end{aligned}$$

From the outer-loop, the inner-current reference commands i_{dr}^* and i_{qr}^* can be expressed as follows :

$$\begin{aligned} i_{dr}^* &= K_{pP_s}(P_s^* - P_s) + K_{iP_s}\Phi_{P_s} \\ i_{qr}^* &= K_{pQ_s}(Q_s - Q_s^*) + K_{iQ_s}\Phi_{Q_s} \quad (9) \end{aligned}$$

Finally, substituting (9) into (8) yields :

$$\begin{aligned} v_{dr} &= K_{pir}(K_{pP_s}(P_s^* - P_s) + K_{iP_s}\Phi_{P_s} - i_{dr}) \\ &\quad + K_{iir}\Phi_{i_{dr}} - \omega_{sl} \left(\sigma X_r i_{qr} - \frac{X_m}{X_s} v_{ds} \right) \quad (10) \end{aligned}$$

$$\begin{aligned} v_{qr} &= K_{pir}(K_{pQ_s}(Q_s - Q_s^*) + K_{iQ_s}\Phi_{Q_s} - i_{qr}) \\ &\quad + K_{iir}\Phi_{i_{qr}} + \omega_{sl} \cdot \sigma X_r i_{dr} \quad (11) \end{aligned}$$

where the controller constants K_{pir} and K_{iir} denote the proportional and integral gains of the inner loop current controller. The constant $\sigma = 1 - X_m/(X_s X_r)$ is known as the leakage coefficient of the induction machine. It is evident that the RSC controller includes cross-coupling terms to compensate for the coupling effects in the dq rotating frame, thereby improving the dynamic performance in tracking control references.

G. GRID-SIDE CONVERTER

Similar to the RSC, the GSC also uses a cascaded control structure to regulate the converter voltage as evident from Fig. 3. This controller operates in a synchronous reference frame rotating at the speed ω_s . The reference frame is aligned with v_s , enabling independent control of the DC-link voltage V_{DC} and the converter reactive power Q_c . The primary role of the GSC is to ensure DC-link voltage stability and provide reactive power support to the grid, thereby enhancing overall system performance and stability. Consequently, the dynamics of the GSC can be described by the following equations:

$$\dot{x}_{gsc} = \mathbf{A}_{gsc}x_{gsc} + \mathbf{B}_{gsc}u_{gsc} \quad (12)$$

$$\mathbf{A}_{gsc} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & K_{iQ_c} & 0 & 0 \\ K_{iv_{DC}} & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{B}_{gsc} = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & K_{pQ_c} & -K_{pQ_c} & -1 & 0 \\ K_{pv_{DC}} & -K_{pv_{DC}} & 0 & 0 & 0 & -1 \end{bmatrix}$$

where $x_{gsc} = [\Phi_{v_{DC}} \ \Phi_{Q_c} \ \Phi_{i_{dc}} \ \Phi_{i_{qc}}]^T$ and $u_{gsc} = [v_{DC}^* \ v_{DC} \ Q_c \ Q_c^* \ i_{dc} \ i_{qc}]^T$. The constants $K_{pv_{DC}}$, $K_{iv_{DC}}$, K_{pQ_c} , and K_{iQ_c} represent the proportional and integral gains of the PI controllers in the GSC. The GSC controller has four states: $\Phi_{v_{DC}}$, Φ_{Q_c} , $\Phi_{i_{dc}}$, and $\Phi_{i_{qc}}$. The GSC voltages v_{dc} and v_{qc} can be calculated in a similar manner to the RSC voltage in (10). Considering the inner-loop controllers, it can be shown that the GSC's voltages v_{dc} and v_{qc} equations are as follows :

$$\begin{aligned} v_{dc} &= K_{pic}(i_{dc}^* - i_{dc}) + K_{iic}\Phi_{i_{dc}} - v_{ds} - \omega_s X_c i_{dc} \\ v_{qc} &= K_{pic}(i_{qc}^* - i_{qc}) + K_{iic}\Phi_{i_{qc}} - v_{qs} + \omega_s X_c i_{qc} \quad (13) \end{aligned}$$

The inner-current reference commands i_{dc}^* and i_{qc}^* can be obtained from the outer-loop controller as follows :

$$\begin{aligned} i_{dc}^* &= K_{pQ_c}(Q_c - Q_c^*) + K_{iQ_c}\Phi_{Q_c} \\ i_{qc}^* &= K_{pv_{DC}}(v_{DC}^* - v_{DC}) + K_{iv_{DC}}\Phi_{v_{DC}} \quad (14) \end{aligned}$$

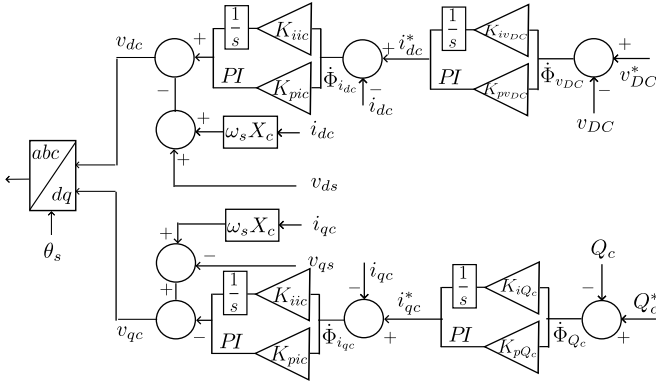


FIGURE 3. Grid-side converter controller.

The final equation can be obtained by a substitution of (14) into (13) as follows:

$$v_{dc} = K_{pic} (K_{pqc} (Q_c - Q_c^*) + K_{iqc} \Phi_{Q_c} - i_{dc}) + K_{iic} \Phi_{i_{dc}} - v_{ds} - \omega_s X_c i_{dc} \quad (15)$$

$$v_{qc} = K_{pic} (K_{pvc} (v_{DC}^* - v_{DC}) + K_{ivc} \Phi_{v_{DC}} - i_{qc}) + K_{iic} \Phi_{i_{qc}} - v_{qs} + \omega_s X_c i_{qc} \quad (16)$$

The controller constants K_{pir} and K_{iir} respectively represent the proportional and integral gains of the inner-loop current controller of the GSC. Additionally, the GSC controller also incorporates cross-coupling terms similar to those used in the RSC to enhance dynamic performance and achieve better current reference tracking.

H. PHASE-LOCKED LOOP

A three-phase PLL is intended to track the angle θ_s and the magnitude of the DFIG's terminal voltage $|v_s|$. This is done by forcing v_{qs} , i.e. the q -axis component of v_s , to zero using a PI controller. The PLL plays a critical role in ensuring the stability of the DFIG system by providing accurate synchronization with the grid voltage $v_s = |v_s| \angle \theta_s$. The dynamic model of this PLL can be expressed as follows:

$$\begin{aligned} \dot{\omega}_{PLL} &= K_{iPLL} v_{qsPLL} \\ \dot{\theta}_{PLL} &= \omega_{PLL} + K_{pPLL} v_{qsPLL} \end{aligned} \quad (17)$$

where $v_{qsPLL} = |v_s| \sin(\theta_s - \theta_{PLL})$ controller constants K_{pPLL} and K_{iPLL} represent the PI controller constants of the PLL, while the PLL states ω_{PLL} and θ_{PLL} correspond to the estimated frequency and phase angle of the terminal voltage v_s .

III. GRID FORMING CONTROLS FOR DFIG

As shown in Fig. 1, the GSC in the GFM mode remains unchanged from the configuration in GFL mode, namely regulating V_{DC} and Q_c . For the RSC, highlighted in the shaded light red area, the inner-loop current controller retains relatively the same structure as in GFL mode. However instead of regulating P_s and Q_s , the outer-loop now controls the stator voltages v_{ds} and v_{qs} . Notably, this control strategy

requires minimal modifications to the inverter controllers, as detailed in [36], enabling the possibility of retrofitting of GFL mode into GFM mode with relatively minimal hardware changes. Additionally, virtual impedance can be seamlessly incorporated into the control structure to improve system stability.

A. OUTER-LOOP VOLTAGE CONTROLLER

The purpose of the voltage controller in GFM mode is to regulate the terminal voltage of the DFIG, enabling grid-forming behavior by directly controlling the inverter voltage magnitude and phase. This is achieved by comparing the measured components v_{ds} and v_{qs} with their respective reference values via PI controllers. In addition, the voltage controller also serves to protect the inverter from over-voltage conditions in particular during faults.

In this paper, the implementation of the voltage controller follows the approach proposed in [36], which allows for minimal hardware modifications from the GFL-mode control structure. Specifically, v_{qs} is regulated to zero, as explained in [36]; it estimates the voltage angle difference between the rotor and the DFIG terminal, i.e., $\theta_r - \theta_s$. This effectively results in the control of the stator active power P_s . Accordingly, the dynamic model of the voltage controller is given as follows:

$$\begin{aligned} \dot{\varphi}_{v_{qs}} &= -v_{qs} - R_{vir} i_{dr} + X_{vir} i_{qr} \\ \dot{\varphi}_{v_{ds}} &= v_{ds} + R_{vir} i_{qr} + X_{vir} i_{dr} - v_{ref} \end{aligned} \quad (18)$$

where $\varphi_{v_{qs}}$ and $\varphi_{v_{ds}}$ are the states of the outer-loop voltage controller. v_{ref} is the reference voltage obtained from the RPC. R_{vir} and X_{vir} represent the resistor and reactance of the virtual impedance Z_{vir} , i.e. $Z_{vir} = R_{vir} + jX_{vir}$.

B. INNER-LOOP CURRENT CONTROLLER

The purpose of the current controller in the GFM mode is the same as in the GFL mode, namely to limit the current flowing through the inverter with the aim to protect the device. The dynamic model of the current controller is as follows:

$$\begin{aligned} \dot{\varphi}_{i_{dr}} &= K_{iPs} \varphi_{v_{qs}} + K_{pPs} (-R_{vir} i_{dr} + X_{vir} i_{qr} - v_{qs}) \\ \dot{\varphi}_{i_{qr}} &= K_{iQs} \varphi_{v_{ds}} + K_{pQs} (v_{ds} - v_{ref} + R_{vir} i_{qr} + X_{vir} i_{dr}) \end{aligned} \quad (19)$$

where $\varphi_{i_{dr}}$ and $\varphi_{i_{qr}}$ denote the states of current controller of RSC GFM. The constants K_{pPs} , K_{iPs} , K_{pQs} and K_{iQs} are the PI controller constants of the voltage regulators. Finally, the rotor voltage can be obtained using the following algebraic equations:

$$v_{dr} = K_{pir} (K_{pPs} (P_s^* - P_s) + K_{iPs} \varphi_{P_s} - i_{qr}) + K_{iir} \varphi_{i_{dr}} - \omega_{sl} \left(\sigma X_r i_{qr} - \frac{X_m}{X_s} v_{ds} \right) \quad (20)$$

$$v_{qr} = K_{pir} (K_{pQs} (Q_s - Q_s^*) + K_{iQs} \varphi_{Q_s} - i_{dr}) + K_{iir} \varphi_{i_{qr}} + \omega_{sl} \cdot \sigma X_r i_{dr} \quad (21)$$

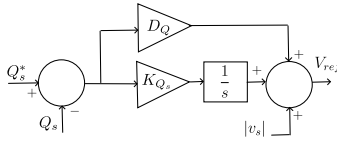


FIGURE 4. Reactive power control.

C. VIRTUAL IMPEDANCE

The implementation of virtual impedance Z_{vir} into the DFIG control structure is given in Fig. 1. As obvious, it is characterized by the parameters R_{vir} and X_{vir} to modify the voltage reference v_{ds}^* and v_{qs}^* signals. This is intended to reshape inverter's impedance such that the system's stability and robustness are improved. Specifically, the virtual impedance is employed to improve power-sharing in microgrids [40], as well as stabilizing GFM inverters under strong grid conditions [41].

In the context of GFM with dual-loop control structures (inner-loop current control and outer-loop voltage control) as given in Fig. 1, it has been shown in [41], [42] that the system may experience stability issues under strong grid conditions, primarily due to low grid inductance and insufficient damping between voltage sources. Whereas, a single-loop with outer-loop voltage control has been demonstrated to improve dynamic performance and system's stability owing to its higher effective reactance. However, without the current controller, there is a risk of over-current flowing through the inverter during faults and hence a promising alternative is to retain the dual-loop architecture while integrating a virtual impedance. Hence, in this paper, the virtual impedance is incorporated into the dual-loop GFM DFIG control structure to improve stability under various grid conditions. The equation for the virtual impedance is given as follows:

$$\begin{aligned} v_s^{aux} &= Z_{vir} i_r \\ &= (R_{vir} + jX_{vir})(i_{dr} + ji_{qr}) \\ &= \underbrace{R_{vir}i_{dr} - X_{vir}i_{qr}}_{v_{ds}^{aux}} + j \underbrace{(R_{vir}i_{qr} + X_{vir}i_{dr})}_{v_{qs}^{aux}} \end{aligned} \quad (22)$$

where $v_s^{aux} = v_{ds}^{aux} + jv_{qs}^{aux}$ denotes the auxiliary voltage references and should be used to subtract v_{ds}^* and v_{qs}^* signals (see Fig. 1).

D. REACTIVE POWER CONTROLLER

Droop control has been widely used in power electronics and microgrid control by emulating the behavior of synchronous generators. This is achieved by adjusting the inverter's voltage magnitude in proportional to its reactive power output as given in Fig 4. The control strategy is relatively simple and enables effective reactive power sharing among parallel inverters without the need for communication. The dynamic equations associated with RPC are given as follows:

$$\dot{Q}_s = k_{Q_s} (Q_s^* - Q_s) \quad (23)$$

$$V_{ref} = |v_s| + D_Q (Q_s^* - Q_s) + \varphi_{Q_s} \quad (24)$$

where the variables Q_s and Q_s^* denote the stator's reactive power and its reference, respectively. The parameter k_{Q_s} represents the integral gain of RPC and is used to eliminate steady-state error in reactive power tracking. The droop coefficient for the reactive power is given by D_Q . It is clear that the inverter reduces its output voltage reference as the reactive power output Q_s increases. As given in Fig. 1 and also (23), the output of RPC is v_{ref} which acts as the q -axis voltage reference for v_s (v_{qs}^*). In the absence of Z_{vir} , v_{ref} is exactly the reference for the DFIG's terminal voltage.

E. ACTIVE POWER CONTROLLER

The active power controller (APC) generates θ_s by measuring the deviation between the stator power P_s and its reference P_s^* . This differs from the GFL DFIG approach, where θ_s is derived using a PLL. By using this method, the behavior of a synchronous generator can be effectively emulated.

1) Droop Control

The schematic diagram of the droop-based APC is shown in Fig.5a. This control method is characterized by the droop coefficient D_P , which governs the dynamic response of the controller. The fundamental principle of the droop-based APC is similar to that of the droop-based RPC; however, the key difference lies in the control objective, namely, the RPC regulates reactive power Q_s , whereas the APC controls the stator active power P_s . Despite this, the droop characteristic is designed to emulate the natural behavior of synchronous generators in sharing active power in response to frequency deviations. The droop gain significantly influences both the transient response and the small-signal stability of the system. A large D_p improves responsiveness but may lead to a condition in which the controller becomes more susceptible to noise and even system's instability. The differential equation governing the droop-based APC is given as follows:

$$\dot{\theta}_s = D_p \omega_b (P_s^* - P_s) \quad (25)$$

2) Virtual Synchronous Generator Control

The control scheme of VSG is given in Fig.5b. At this point, both droop and VSG aim to emulate synchronous generator behavior to enable grid-forming capability in inverters, the detail implementation of both the algorithms has few technical differences. The droop control adjusts frequency and voltage magnitude in proportion to active and reactive power output, based on steady-state relationships, whereas VSG incorporates dynamic equations, specifically the swing equation of synchronous generators, to emulate rotor swing dynamics. With this approach, VSG is able to provide synthetic inertia, allowing inverter-based resources to contribute to system frequency support similar to conventional generators. Droop controller, on the other hand, does not naturally provide enough inertia, leading to large RoCoF particularly during standalone mode.

Further, VSG implementations may vary slightly among different algorithm. For example, fixed values of J and D_p

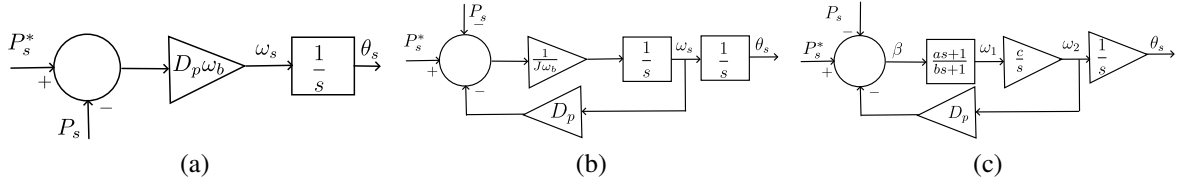


FIGURE 5. Active power controller. (a) Droop, (b) VSG, and (c) GVSG.

have been used in [43]. To improve the performance, an adaptive VGS in which J and D_p are tuned dynamically is proposed in [44]. More recently, a robust control approach considering parameter uncertainty has been used in [45] to design the VSG parameters. Finally, the dynamic model of VSG-based RSC GFM is given as follows:

$$\dot{x}_{vsg} = \mathbf{A}_{vsg}x_{vsg} + \mathbf{B}_{vsg}u_{vsg} \quad (26)$$

$$\mathbf{A}_{vsg} = \begin{bmatrix} -D_p & 0 \\ 1 & 0 \end{bmatrix}$$

$$\mathbf{B}_{vsg} = \begin{bmatrix} 1/J\omega_b & -1/J\omega_b \\ 0 & 0 \end{bmatrix}$$

where $x_{vsg} = [\omega_s \ \theta_s]^T$ and $u_{vsg} = [P_s^* \ P_s]^T$. J and D_p are the inertia constant and the damping coefficient of VSG, respectively. It is apparent that (26) is exactly the swing equation of synchronous generator's rotor.

3) Generalized Virtual Synchronous Generator Control

GVSG was initially introduced in [9] with the aim of enhancing the dynamic performance of conventional VSG control. Specifically, it seeks to improve both active power reference tracking and virtual inertia support by incorporating a lead-lag compensator into the control loop, as illustrated in Fig.5c. To this end, the GVSG parameters are selected through an analytically defined procedure that offers flexibility in shaping the system's transient response. The design process begins by defining the following intermediate constants:

$$\alpha = \tau, \ \beta = \tau^2 \sqrt{\frac{k_g^2 D_p^2}{\tau} - 1}, \ \gamma = \frac{1}{\sqrt[3]{k_g^2 D_p^2 / \tau_i^p - 1}} \quad (27)$$

where τ is a constant associated with RoCoF and is set to 0.02 in this paper (see [9] for more detail). $k_g = \frac{3|v_s||v_g|}{\omega_b L_T}$ with $L_T = \Im(Z_T)/2\pi$, namely the inductive element of Z_T . These intermediate constants are used to compute the final controller gains as follows:

$$a = \alpha, \ b = \beta\gamma/(\beta + \gamma - \alpha), \ c = (\beta + \gamma - \alpha)/D_p \quad (28)$$

Finally, according to Fig.5c, the dynamic equations governing GVSG are:

$$\dot{x}_{gvsg} = \mathbf{A}_{gvsg}x_{gvsg} + \mathbf{B}_{gvsg}u_{gvsg} \quad (29)$$

$$\mathbf{A}_{gvsg} = \begin{bmatrix} -1/b & -D_p/b & 0 \\ c(b-a)/b & -caD_p/b & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{B}_{gvsg} = \begin{bmatrix} 1/b & -1/b \\ ca/b & -ca/b \\ 0 & 0 \end{bmatrix}$$

The variables $x_{gvsg} = [\rho \ \omega_2 \ \theta_s]^T$ and $u_{gvsg} = [P_s^* \ P_s]^T$ denote the states of internal dynamic and inputs of GVSG. Note that if $a = b = c = J\omega_b$, then GVSG becomes VSG.

IV. NETWORK MODEL

As illustrated in Fig. 1, the remaining components of the system include the dynamics of the terminal voltage, the transformer, the grid impedance, and the load. This section introduces the differential equations that govern these components.

A. TERMINAL VOLTAGE

To capture the dynamics of the terminal voltage $v_s = v_{ds} + jv_{qs}$, a shunt susceptance B_s is introduced. From Fig. 1, the dynamic of v_s is influenced by the balance between the current injected by the DFIG, denoted as $i_{B_s} = i_{dB_s} + ji_{qB_s}$, and consisting of the sum between $i_s = i_{ds} + ji_{qs}$ and $i_c = i_{dc} + ji_{qc}$, and the series-compensated line current, $i_l = i_{dl} + ji_{ql}$, can be expressed as $i_{B_s} = (i_s - i_g) - i_l$. Consequently, from the basic circuit theory, v_s is described as follows:

$$\dot{x}_{sh} = \mathbf{A}_{sh}x_{sh} + \mathbf{B}_{sh}u_{sh} \quad (30)$$

$$\mathbf{A}_{sh} = \begin{bmatrix} 0 & -\omega_b\omega_s \\ \omega_b\omega_s & 0 \end{bmatrix}$$

$$\mathbf{B}_{sh} = \begin{bmatrix} \omega_b/B_s & 0 \\ 0 & \omega_b/B_s \end{bmatrix}$$

where the state and input vectors are defined as $x_{sh} = [v_{qs} \ v_{ds}]^T$ and $u_{sh} = [i_{qB_s} \ i_{dB_s}]^T$, respectively.

B. TRANSFORMER

A transformer steps up the DFIG terminal voltage to minimize transmission losses and enable connection to the point of common coupling (PCC). It plays a key role in determining the power flow and dynamic interaction with the grid. Its dynamic behavior can be modeled using a series RL circuit expressed in the synchronous dq reference frame as follows:

$$\dot{x}_{tr} = \mathbf{A}_{tr}x_{tr} + \mathbf{B}_{tr}u_{tr} \quad (31)$$

$$\mathbf{A}_{tr} = \begin{bmatrix} -\omega_b R_T/L_T & -\omega_b\omega_s \\ \omega_b\omega_s & -\omega_b R_T/L_T \end{bmatrix}$$

$$\mathbf{B}_{tr} = \begin{bmatrix} \omega_b/L_T & 0 \\ 0 & \omega_b/L_T \end{bmatrix}$$

where $x_{tr} = [i_{qt} \ i_{dt}]^T$ and $u_{tr} = [v_{qs} - v_{qpcc} \ v_{ds} - v_{dpcc}]^T$. The state variables i_{dt} and i_{qt} are the dq -component of transformer current i_t . R_T and L_T illustrate the resistance and inductance of series RL element, respectively.

C. LOAD MODEL

The load is assumed to be connected between the PCC and ground. In this study, the load $S_L = P_L + jQ_L$ is modeled as a constant impedance. The corresponding equivalent impedance is calculated as follows:

$$Z_L = |v_{pcc}|^2/S^* = |v_{pcc}|^2/(P_L - jQ_L) \quad (32)$$

Once the values of $R_L = \Re(Z_L)$ and $L_L = \Im(Z_L)$ are extracted from Z_L , they are used in the following differential equations to describe the dynamic behavior of the load current:

$$\begin{aligned} \dot{x}_L &= \mathbf{A}_L x_L + \mathbf{B}_L u_L \\ \mathbf{A}_L &= \begin{bmatrix} -\omega_b/L_L R_L & -\omega_b \omega_s \\ \omega_b \omega_s & -\omega_b/L_L R_L \end{bmatrix} \\ \mathbf{B}_L &= \begin{bmatrix} \omega_b/L_L & 0 \\ 0 & \omega_b/L_L \end{bmatrix} \end{aligned} \quad (33)$$

where $x_L = [i_{qL} \ i_{dL}]^T$ and $u_L = [v_{qpcc} \ v_{dpcc}]^T$. i_{dL} and i_{qL} represent the state variables of d and q -components of the load current i_L , respectively. The parameters R_L and L_L are the resistance and inductance of the equivalent load impedance, respectively.

D. GRID IMPEDANCE

The dynamic model of the grid impedance closely resembles that of the transformer, as both are represented using a series RL configuration in the dq reference frame. Importantly, the value of L_g reflects the grid strength—stronger grids are typically characterized by lower impedance values, while weaker grids exhibit higher impedance. The corresponding differential equations that describe the dynamic behavior of the grid impedance are given as follows:

$$\begin{aligned} \dot{x}_g &= \mathbf{A}_g x_g + \mathbf{B}_g u_g \\ \mathbf{A}_g &= \begin{bmatrix} -\omega_b R_g/L_g & -\omega_b \omega_s \\ \omega_b \omega_s & -\omega_b R_g/L_g \end{bmatrix} \\ \mathbf{B}_g &= \begin{bmatrix} \omega_b/L_g & 0 \\ 0 & \omega_b/L_g \end{bmatrix} \end{aligned} \quad (34)$$

The state and input vectors are defined as $x_g = [i_{qg} \ i_{dg}]^T$ and $u_g = [v_{qpcc} - v_{qg} \ v_{dpcc} - v_{dg}]^T$, respectively. The quantities i_{dg} and i_{qg} represent the dq -axis components of the current passing through the grid-side impedance. The terms R_g and L_g model the resistance and inductance characterizing the grid's electrical behavior.

V. MODAL ANALYSIS

The modal analysis is done through linearization of the differential-algebraic equations of the system, as illustrated in the following two subsections.

A. DIFFERENTIAL-ALGEBRAIC EQUATIONS

Sections II, III, and IV presented the full set of dynamic equations for the GFL inverter, GFM inverter, and the network components, respectively. These equations can be collectively expressed in a compact form as a set of differential-algebraic equations (DAEs), as shown by [46], [47]:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{z}, \mathbf{u}), \quad \mathbf{0} = \mathbf{g}(\mathbf{x}, \mathbf{z}, \mathbf{u}), \quad \mathbf{y} = \mathbf{h}(\mathbf{x}, \mathbf{z}, \mathbf{u}) \quad (35)$$

In (35), the functions $\mathbf{f}(\cdot)$, $\mathbf{g}(\cdot)$, and $\mathbf{h}(\cdot)$ respectively represent the set of nonlinear differential, algebraic, and the output equations governing the dynamics of the system in Fig. 1. The variables $\mathbf{x}$, $\mathbf{z}$, $\mathbf{u}$, and $\mathbf{y}$ are state, algebraic, input, and output variables, respectively.

To clarify the structure of the nonlinear DAE model of (35) for GFL and GFM systems employing droop, VSG, and GVSG control strategies, the corresponding definitions of the state vector $\mathbf{x}$, algebraic vector $\mathbf{z}$, and input vector $\mathbf{u}$ are discussed for each case. In the case of GFL, the vectors representing the DAE (35) are given as follows:

$$\begin{aligned} \mathbf{x}_{\text{GFL}} &= [i_{qs}, i_{ds}, i_{qr}, i_{dr}, \omega_t, \omega_r, T_g, i_{qc}, i_{dc}, v_{DC}, \Phi_{idr}, \\ &\quad \Phi_{iqr}, \Phi_{Ps}, \Phi_{Qs}, \Phi_{v_{DC}}, \Phi_{Qc}, \Phi_{ide}, \Phi_{iqc}, \omega_{PLL}, \\ &\quad \theta_{PLL}, v_{ds}, v_{qs}, i_{qt}, i_{dt}, i_{qL}, i_{dL}, i_{qg}, i_{dg}]^T \\ \mathbf{u}_{\text{GFL}} &= [v_w, Q_s^*, v_{DC}^*, Q_c^*]^T \\ \mathbf{z}_{\text{GFL}} &= [T_m, P_r, P_c, P_s^*, v_{dr}, v_{qr}, v_{dc}, v_{qc}]^T \end{aligned} \quad (36)$$

The vectors for GFM with droop controller are expressed as follows:

$$\begin{aligned} \mathbf{x}_{\text{DRP}} &= [i_{qs}, i_{ds}, i_{qr}, i_{dr}, \omega_t, \omega_r, T_g, i_{qc}, i_{dc}, v_{DC}, \\ &\quad \varphi_{idr}, \varphi_{iqr}, \varphi_{v_{qs}}, \varphi_{v_{qs}}, \varphi_{v_{DC}}, \varphi_{Qc}, \varphi_{ide}, \varphi_{iqc}, \\ &\quad \theta, \varphi_{Qs}, v_{ds}, v_{qs}, i_{qt}, i_{dt}, i_{qL}, i_{dL}, i_{qg}, i_{dg}, \\ &\quad \omega_{PLL}, \theta_{PLL}]^T \\ \mathbf{u}_{\text{DRP}} &= [v_w, Q_s^*, v_{DC}^*, Q_c^*]^T \\ \mathbf{z}_{\text{DRP}} &= [T_m, P_r, P_c, P_s^*, V_{ref}, v_{ds}^{aux}, v_{qs}^{aux}, \\ &\quad v_{dr}, v_{qr}, v_{dc}, v_{qc}]^T \end{aligned} \quad (37)$$

The vectors for GFM with VSG control strategy are given as follows:

$$\begin{aligned} \mathbf{x}_{\text{VSG}} &= [i_{qs}, i_{ds}, i_{qr}, i_{dr}, \omega_t, \omega_r, T_g, i_{qc}, i_{dc}, v_{DC}, \\ &\quad \varphi_{idr}, \varphi_{iqr}, \varphi_{v_{qs}}, \varphi_{v_{qs}}, \varphi_{v_{DC}}, \varphi_{Qc}, \varphi_{ide}, \varphi_{iqc}, \\ &\quad \theta, \omega, \varphi_{Qs}, v_{ds}, v_{qs}, i_{qt}, i_{dt}, i_{qL}, i_{dL}, i_{qg}, i_{dg}, \\ &\quad \omega_{PLL}, \theta_{PLL}]^T \\ \mathbf{u}_{\text{VSG}} &= [v_w, Q_s^*, v_{DC}^*, Q_c^*]^T \\ \mathbf{z}_{\text{VSG}} &= [T_m, P_r, P_c, P_s^*, V_{ref}, v_{ds}^{aux}, v_{qs}^{aux}, \\ &\quad v_{dr}, v_{qr}, v_{dc}, v_{qc}]^T \end{aligned} \quad (38)$$

Finally, for the case of GVSG, the corresponding vectors are given in as follows:

$$\begin{aligned} \mathbf{x}_{CVSG} &= [i_{qs}, i_{ds}, i_{qr}, i_{dr}, \omega_t, \omega_r, T_g, i_{qc}, i_{dc}, v_{DC} \\ &\quad \varphi_{i_{dr}}, \varphi_{i_{qr}}, \varphi_{v_{qs}}, \varphi_{v_{qs}}, \varphi_{v_{DC}}, \varphi_{Q_c}, \varphi_{i_{dc}} \\ &\quad \varphi_{i_{qc}}, \rho, \omega_2, \theta_s, \varphi_{Q_s}, v_{ds}, v_{qs}, i_{qt}, i_{dt} \\ &\quad i_{qL}, i_{dL}, i_{qg}, i_{dg}, \omega_{PLL}, \theta_{PLL}]^T \\ \mathbf{u}_{CVSG} &= [v_w, Q_s^*, v_{DC}^*, Q_c^*]^T \\ \mathbf{z}_{CVSG} &= [T_m, P_r, P_c, P_s^*, V_{ref}, v_{ds}^{aux}, v_{qs}^{aux} \\ &\quad v_{dr}, v_{qr}, v_{dc}, v_{qc}]^T \end{aligned} \quad (39)$$

B. LINEARIZATION

In the small-signal stability analysis, the nonlinear system in (35) is linearized around an equilibrium point $(\mathbf{x}_0, \mathbf{z}_0, \mathbf{u}_0)$. This point is the solution to (35) with the derivative term $\dot{\mathbf{x}}$ set to zero. In this paper, a two-stage approach is used to calculate $(\mathbf{x}_0, \mathbf{z}_0, \mathbf{u}_0)$:

- 1) *Stage-1: Load flow.* The load flow problem is solved using the Newton-Raphson method. Once converged, the network states such as $v_{ds}, v_{qs}, i_{ql}, i_{dl}$ can be easily calculated using basic circuit theories such as Kirchhoff's laws.
- 2) *Stage-2: Machine initialization.* With the load flow calculation in Stage-1, the terminal variables such as V_s and θ_s are known. These variables are subsequently used to initialize DFIG states by setting the derivative term $\dot{\mathbf{x}}$ in (35) to zero. In this step, Levenberg-Marquardt algorithm is deployed.

Once the equilibrium point is identified, the linearization can be performed. In this paper, numerical linearization method using Matlab's function *linearize* is pursued. By eliminating the algebraic variable $\mathbf{z}$, the standard linearized state-space model is given as follows:

$$\Delta \dot{\mathbf{x}} = A\Delta \mathbf{x} + B\Delta \mathbf{u}, \text{ and } \Delta \mathbf{y} = C\Delta \mathbf{x} + D\Delta \mathbf{u} \quad (40)$$

where the matrices A , B , C and D are calculated using the relevant partial derivatives of the functions $f(\cdot)$, $g(\cdot)$, and $h(\cdot)$ with respect to the variables $\mathbf{x}$, $\mathbf{z}$, $\mathbf{u}$ as follows:

$$\begin{aligned} A &= \frac{\partial f}{\partial \mathbf{x}} - \frac{\partial f}{\partial \mathbf{z}} \left(\frac{\partial g}{\partial \mathbf{z}} \right)^{-1} \frac{\partial g}{\partial \mathbf{x}}, \quad B = \frac{\partial f}{\partial \mathbf{u}} - \frac{\partial f}{\partial \mathbf{z}} \left(\frac{\partial g}{\partial \mathbf{z}} \right)^{-1} \frac{\partial g}{\partial \mathbf{u}} \\ C &= \frac{\partial h}{\partial \mathbf{x}} - \frac{\partial h}{\partial \mathbf{z}} \left(\frac{\partial g}{\partial \mathbf{z}} \right)^{-1} \frac{\partial g}{\partial \mathbf{x}}, \quad D = \frac{\partial h}{\partial \mathbf{u}} - \frac{\partial h}{\partial \mathbf{z}} \left(\frac{\partial g}{\partial \mathbf{z}} \right)^{-1} \frac{\partial g}{\partial \mathbf{u}} \end{aligned} \quad (41)$$

The eigenvalue $\lambda_i = \sigma_i \pm j\omega_i$ along with the corresponding left eigenvector l_i and right eigenvector r_i of the linearized system matrix A is defined as follows:

$$A r_i = \lambda_i r_i \text{ and } l_i^T A = \lambda_i l_i^T \quad (42)$$

The system is stable if the real parts of all eigenvalues λ_i are negative, namely $\sigma_i < 0$. The corresponding damping

ratio and oscillation frequency of each eigenvalue can be determined using the following expressions:

$$\zeta_i = -\sigma_i / \sqrt{\sigma_i^2 + \omega_i^2} \text{ and } f_i = \omega_i / 2\pi \quad (43)$$

In power system studies, it is often beneficial to identify the dominant state variables associated with a particular λ_i . To do this, the participation factors are used. The participation vector for the i^{th} mode, which quantifies the involvement of each of the n state variables, is given by:

$$\mathbf{p}_i = [p_{1i}, p_{2i}, \dots, p_{ni}]^T \quad (44)$$

Each component p_{ki} represents the normalized contribution of the k^{th} state to the i^{th} mode, calculated as:

$$p_{ki} = |\ell_{ik}| |r_{ki}| / \sum_{k=1}^n |\ell_{ik}| |r_{ki}| \quad (45)$$

Here, ℓ_{ik} and r_{ki} denote the k^{th} components of the left and right eigenvectors associated with the i^{th} mode, respectively. These participation factors provide insight into which states have the greatest influence on each dynamic mode and are widely used in modal analysis to identify critical modes and their associated dynamic states. With the formula in (45), the state that contributes most significantly to a given mode λ_i will have a unity participation factor.

VI. TEST SYSTEM

The test system utilized in this study is shown in Fig. 1 and is adapted from the model presented in [38]. It comprises a 2 MW DFIG delivering power to the grid via a step-up transformer. The transformer increases the generator terminal voltage from 690 V to 161 kV to match the transmission level. The grid is represented as an infinite bus connected through a series impedance, the value of which defines the grid strength. A load modeled as a constant impedance load is assumed to be installed at the PCC. Note that the series compensator has been removed from the original paper in [38] as the focus of the current study is not on SSR oscillations.

A basic operating condition is assumed when the DFIG delivers 1 MW of active power (0.5 pu) to the grid with unity power factor. The per unit voltages for generator terminal v_s , PCC v_{pcc} , and grid voltage v_g are set to $1.01 \angle 18.72^\circ$, $0.97 \angle 4.41^\circ$, and $1.00 \angle 0.0^\circ$, respectively. The load S_L consumes $0.2 + j0.1$. Further, the infinite bus absorbs active power 0.28 pu, while delivers reactive power 0.14 pu to the load. With this load flow results, the equilibrium point can be calculated using the method given in Subsection V-B. An example of the initial states for the DFIG with droop control is provided in Table 1, which represents the operating point at which the system is linearized. Control strategies such as GFL or VSG can be developed following a similar approach; however, they are not presented here due to space constraints.

Finally, Table 2 summarizes all relevant parameters used throughout this paper, which are adopted from [17]. These include the electrical characteristics of the grid and network,

TABLE 1. Initial States for Droop Controller

State	Value	State	Value	State	Value	State	Value
v_{DC}	1200	i_{dc}	0.12	i_{qc}	0.04	$\Phi_{i_{dc}}$	0.01
$\Phi_{i_{qc}}$	0.00	Φ_{Q_c}	0.00	$\Phi_{v_{DC}}$	0.13	ω_{PLL}	0.00
θ_{PLL}	0.33	ω_t	0.80	ω_r	0.80	T_g	4.23
i_{ds}	0.33	i_{qs}	-0.97	i_{dr}	0.45	i_{qr}	-0.95
θ	0.51	$\Phi_{i_{dr}}$	0.00	$\Phi_{i_{qr}}$	-0.04	$\varphi_{v_{qs}}$	-0.37
$\varphi_{v_{qs}}$	0.58	φ_{Q_s}	0.14	i_{dt}	0.50	i_{qt}	0.06
i_{dL}	0.21	i_{qL}	-0.09	i_{dg}	0.28	i_{qg}	0.15
v_{ds}	0.33	v_{qs}	0.96				

TABLE 2. System Parameters

Symbol	Value	Unit	Symbol	Value	Unit
Grid and network					
v_g	161	kV	ω_b	314	rad/s
S_B	2	MVA	B_s	0.1	pu
R_g	0.025	pu	X_g	0.25	pu
R_T	0.05	pu	X_T	0.5	pu
R_L	0.2	pu	Q_L	0.1	pu
DFIG					
S_{rated}	2.0	MVA	v_s	690	V
X_{ls}	0.092	pu	X_m	3.95279	pu
X_{lr}	0.099	pu	R_s	0.00488	pu
R_r	0.0055	pu	C_{DC}	14×10^{-3}	μF
v_{DC}	1200	V	H_g	0.9	s
H_t	4.29	s	D_t	0.0	pu
D_g	0.0	pu	D_{Tg}	1.5	pu
K_{Tg}	0.15	pu	R	25.32	m
X_c	0.3	pu			
Voltage and current control parameters					
K_{pP_s}	0	pu	K_{iP_s}	3	-
K_{pQ_s}	0.11	-	K_{iQ_s}	11.37	-
K_{pir}	0.3	-	K_{iir}	20	-
$K_{pv_{DC}}$	1.01	-	$K_{iv_{DC}}$	25.5	-
K_{pQ_c}	0	-	K_{pQ_c}	30	-
K_{pic}	0.3	-	K_{iic}	20	-
GFL					
K_{pPLL}	60	-	K_{iPLL}	1400	-
Droop					
R_{vir}	0.105	pu	X_{vir}	0.323	pu
k_{Q_s}	1.2	-	D_Q	0.3	V/pu
D_P	0.03	rad/pu s	n	0.1	pu/kV
VSG					
R_{vir}	0.105	pu	X_{vir}	0.323	pu
k_{Q_s}	1.2	-	D_Q	0.3	V/pu
J	81.04	Ws ² /rad	D	2.5×10^5	Ws/rad
GVSG					
R_{vir}	0.105	pu	X_{vir}	0.323	pu
k_{Q_s}	1.2	-	D_Q	0.3	V/pu
τ	0.02	s	D	2.5×10^5	Ws/rad

detailed parameters of the DFIG model, as well as the control gains for all strategies, namely GFL, droop, VSG, and GVSG. An important observation is that the voltage and current control parameters are assigned the same numerical values for all scenarios, leading to only slight modifications in the respective APC control strategies.

Next the models described in Sections II–IV are implemented in Matlab/Simulink. To validate the accuracy of the time-domain simulation, the proposed model is compared against the built-in model from *SimPowerSystems* for a 32 second simulation. At $t = 29$ s, the system is subjected to a 10% step increase in the infinite bus voltage v_g . Note that the fault is applied near the end of the simulation to allow

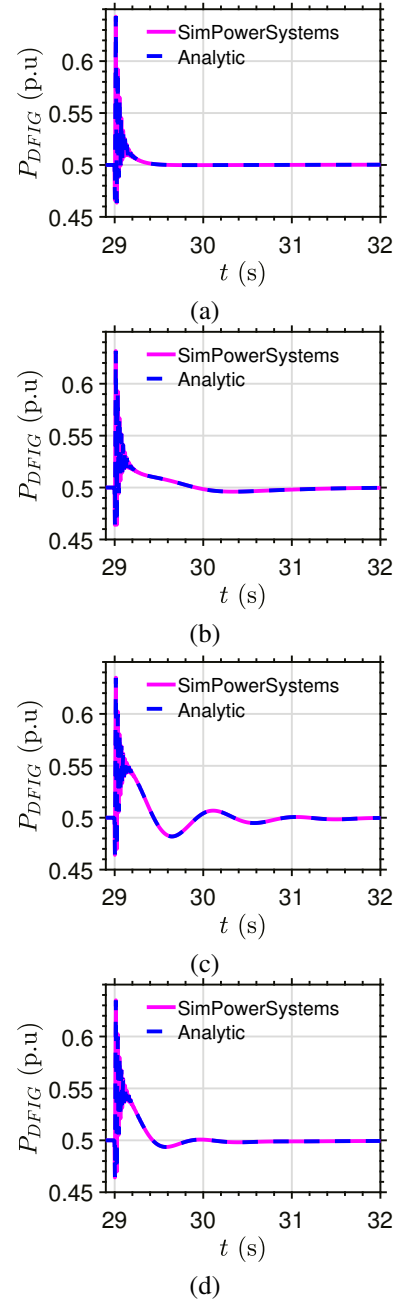


FIGURE 6. Comparison of P_{DFIG} under different control strategies. (a) GFL, (b) Droop, (c) VSG, and (d) GVSG.

the built-in model to reach its equilibrium point before the disturbance occurs. The active power responses of the DFIG (P_{DFIG}) under all control strategies are presented in Fig. 6. As shown, the waveforms from the analytical model match those obtained from *SimPowerSystems* with a high degree of accuracy, confirming the accuracy of the proposed model.

VII. SMALL-SIGNAL ANALYSIS

This section presents a comprehensive small-signal stability analysis of the GFM DFIG system depicted in Fig. 1. The main objective is to investigate the influence of key factors,

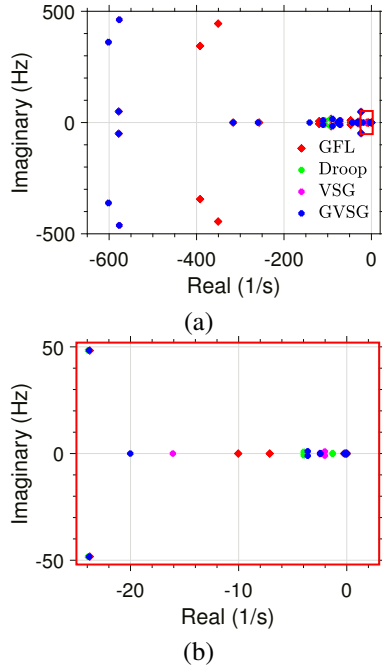


FIGURE 7. Eigenvalue plots for all control strategies. (a) Complete Plot and (b) Zoomed-in Plot.

such as GFM control strategies, inner-current control loops, virtual impedance, and so on, on dominant modes.

A. MODAL ANALYSIS OF GFM CONTROLS

In this scenario, the oscillatory behavior of the GFM-based DFIG system under three different control strategies is thoroughly investigated. The objectives of this analysis are twofold: first, to gain deeper insight into the various modes that exist within the DFIG system, and second, to identify the dominant dynamic behaviors that influence overall system stability. The upcoming sections will focus mostly on these dominant modes in much more detail as different scenarios are analyzed. For the sake of completeness, a comparison with the DFIG with GFL control mode is also provided. The corresponding eigenvalue plots for all control strategies are presented in Fig. 7. A zoomed-in plot is included to clearly highlight the dominant modes. The numerical values for all eigenvalues with the respective frequencies, damping, and participation analysis are given in Table 3. Note that this table only displays the states whose participation factors are larger than 0.7.

It is evident that some modes are only marginally influenced by the choice of control strategy. For instance, the modes λ_1 and λ_2 , which exhibit frequencies above 300 Hz, remain relatively unchanged across all control implementations. Based on participation factor analysis, these high-frequency modes are primarily associated with the dynamics of the induction generator and terminal voltage states, including v_{ds} , v_{qs} , i_{ds} , and i_{qs} . Similarly, the modes near the fundamental frequency (around 50 Hz), specifically λ_3 and λ_4 , also show minimal variation to different control

choices, as they are predominantly linked to network current dynamics such as i_{dt} and i_{qt} . Another key observation is that the modes related to GSC controller and DC-link dynamics, e.g. $\Phi_{v_{DC}}$ or $\varphi_{v_{DC}}$ and v_{DC} , undergo negligible change across all control configurations as the same GSC control structure is utilized. For example, consider λ_{11} in the case of GFL control, or λ_{12} under the VSG strategy.

Despite the fact that some modes are relatively constant, several others are strongly affected by the choice of control strategy. As expected, the most notable differences are observed in the modes associated with RSC states as different control strategies are deployed in the RSC. For example the modes related to RSC in the GFL controller are $\lambda_9 = -47.3 \pm j60.8$, $\lambda_{14} = -10.0$, and $\lambda_{15} = -7.1$, whereas the RSC-related modes are $\lambda_9 = -71.9 \pm j55.4$, $\lambda_{14} = -4.0 \pm j5.1$ and $\lambda_{15} = -1.3 \pm j1.4$ in the droop control strategy, and $\lambda_9 = -71.9 \pm j55.4$, $\lambda_{14} = -2.0 \pm j5.9$, and $\lambda_{15} = -2.4 \pm j1.1$ in the VSG. The respective modes in the GVSG controller are $\lambda_{10} = -72.1 \pm j59.6$, $\lambda_{16} = -3.6 \pm j6.6$ and $\lambda_{17} = -2.4 \pm j1.1$.

Finally, the identification of dominant modes can be performed using the eigenvalue plot in Fig. 7, particularly by examining the zoomed-in region. As observed, the modes with the frequency of 48.3 Hz and damping 7.8 are located close to the $j\omega$ -axis, indicating low damping. As previously discussed, these modes are associated with the network dynamics. Their activity is clearly visible in the time-domain response shown in Fig. 6, especially the fault event.

Interestingly, both the VSG and GVSG controllers exhibit noticeable low-frequency oscillations, despite being well-damped. These oscillations are linked to APC modes. According to Table 3, these correspond to λ_{14} in the VSG case and λ_{16} in GVSG. The APC mode in GVSG demonstrates better damping compared to that in VSG, which can be attributed to the control design of GVSG that enhances the dynamic response of the VSG architecture.

B. IMPACTS OF GRID STRENGTH

This scenario investigates how grid strength influences the stability of a DFIG operating under various control modes. The short-circuit ratio (SCR) is varied from a weak grid condition ($SCR = 2$) to a strong grid condition ($SCR = 16$), while maintaining a constant grid impedance ratio of $X_g/R_g = 10$. The trajectory of eigenvalues is shown in Fig. 8. In general, most eigenvalues shift toward the left half of the complex plane, indicating improved damping. However, a few modes, particularly under GFM control strategies, exhibit movement toward the right, which may indicate potential stability concerns under certain conditions.

Focusing on the eigenvalues with real parts greater than 40 in Fig. 8, it is observed that almost all modes exhibit improved damping as the grid strength increases. Specifically, this includes the RSC inner-loop controller dynamics, i_{qr} and $\Phi_{i_{qr}}$ in GFL (indicated by letter G) or i_{qr} and $\varphi_{i_{qr}}$ in GFM (indicated by letter H), the DC-link dynamics, i_{dc} , v_{DC} , and i_{dc} (indicated by letter I), and the GSC inner-loop dynamics

TABLE 3. Eigenvalue and Participation Factor for All Control Strategies.

No	Eigenvalues λ_i	f_i	ζ_i	States & p_{ki}	Eigenvalues λ_i	f_i	ζ_i	States & p_{ki}
GFL					VSG			
1	$-350.7 \pm j2794.6$	444.8	12.5	v_{qs} (1.0), i_{qs} (0.81)	$-576.9 \pm j2903.1$	462.0	19.5	v_{qs} (1.00), i_{qs} (0.79), v_{ds} (0.75)
2	$-391.9 \pm j2163.8$	344.4	17.8	v_{ds} (1.0), i_{ds} (0.96)	$-601.2 \pm j2271.5$	361.5	25.5	v_{ds} (1.00), i_{ds} (0.91), v_{qs} (0.73)
3	$-578.0 \pm j312.3$	49.7	88.0	i_{dL} (1.0), i_{qL} (0.99)	$-578.1 \pm j312.1$	49.6	87.9	i_{qL} (1.00), i_{dL} (1.00)
4	$-23.8 \pm j303.3$	48.3	7.8	i_{dt} (1.0), i_{qt} (0.98), i_{qs} (0.87), i_{ds} (0.82)	$-23.9 \pm j303.8$	48.3	7.8	i_{qt} (1.00), i_{dt} (0.99), i_{ds} (0.85), i_{qs} (0.81)
5	-316.4	0.0	100.0	ω_r (1.0)	-316.2	0	100	ω_r (1.00)
6	-256.9	0.0	100.0	i_{qc} (1.0)	-259.0	0	100	i_{qc} (1.00)
7	$-92.3 \pm j106.5$	16.9	65.5	v_{DC} (1.0), i_{dr} (0.94), i_{dc} (0.76)	$-89.2 \pm j99.7$	15.8	66.7	v_{DC} (1.00), i_{dc} (0.73), i_{dr} (0.80)
8	$-119.5 \pm j38.6$	6.1	95.2	i_{dc} (1.0), $\Phi_{i_{dc}}$ (0.7)	$-109.6 \pm j59.4$	9.4	87.9	i_{dc} (1.00), v_{DC} (0.78), $\varphi_{i_{dc}}$ (0.76)
9	$-47.3 \pm j60.8$	9.7	61.4	i_{qr} (1.0), $\varphi_{i_{qr}}$ (0.93)	$-71.9 \pm j55.4$	8.8	79.2	i_{qr} (1.00), $\varphi_{i_{qr}}$ (0.95)
10	-76.6	0.0	100.0	$\varphi_{i_{qc}}$ (1.0)	-77.1	0	100	$\varphi_{i_{qc}}$ (1.00)
11	$-45.9 \pm j9.1$	1.5	98.1	$\Phi_{v_{DC}}$ (1.0), v_{DC} (0.9)	$-29.9 \pm j22.7$	3.6	79.6	θ_{PLL} (1.00), ω_{PLL} (0.99)
12	$-27.2 \pm j23.6$	3.8	75.6	θ_{PLL} (1.0), ω_{PLL} (0.97)	$-43.8 \pm j6.1$	0.98	99.0	$\varphi_{v_{DC}}$ (1.00), v_{DC} (0.78)
13	-32.0	0.0	100.0	φ_{Q_c} (1.0)	-32.8	0.0	100	φ_{Q_c} (1.00)
14	-10.0	0.0	100.0	φ_{Q_s} (1.0), φ_{P_s} (0.96)	$-2.0 \pm j5.9$	0.95	31.1	θ_s (1.00)
15	-7.1	0.0	100.0	φ_{P_s} (1.0), φ_{Q_s} (0.92)	$-2.4 \pm j1.1$	0.17	91.5	φ_{Q_s} (1.00), $\varphi_{v_{ds}}$ (0.72), i_{qt} (0.72), $i_{qg} = 0.71$
16	-0.2	0.0	100.0	ω_t (1.0)	-0.23	0	100	ω_t (1.00)
17	-0.1	0.0	100.0	T_g (1.0)	-0.1	0	100	T_g (1.00)
18	-0.0	0	100.0	i_{dg} (1.0), i_{qg} (1.0)	-0.0	0	100	i_{qg} (1.00), i_{dg} (1.00)
19					-16.1	0	100	ω_s (1.00)
Droop					GVSG			
1	$-576.9 \pm j2903.1$	462.0	19.5	v_{qs} (1.00), i_{qs} (0.79), v_{ds} (0.75)	$-576.8 \pm j2903.7$	462.1	19.5	v_{qs} (1.00), i_{qs} (0.79), v_{ds} (0.76)
2	$-601.2 \pm j2271.5$	361.5	25.5	v_{ds} (1.00), i_{ds} (0.91), v_{qs} (0.73)	$-601.5 \pm j2270.5$	361.4	25.6	v_{ds} (1.00), i_{ds} (0.91), v_{qs} (0.73)
3	$-578.1 \pm j312.1$	49.6	87.9	i_{qL} (1.00), i_{dL} (1.00)	$-578.1 \pm j312.1$	49.7	88.0	i_{qL} (1.00), i_{dL} (1.00)
4	$-23.9 \pm j303.8$	48.3	7.8	i_{qt} (1.00), i_{dt} (0.99), i_{ds} (0.85), i_{qs} (0.81)	$-23.8 \pm j303.7$	48.3	7.8	i_{dt} (1.00), i_{qt} (1.00), i_{ds} (0.85), i_{qs} (0.83)
5	-316.2	0	100	ω_r (1.00)	-316.2	0.0	100.0	ω_r (1.00)
6	-259.0	0	100	i_{qc} (1.00)	-259.1	0.0	100.0	i_{qc} (1.00)
7	$-93.5 \pm j98.4$	15.6	68.8	v_{DC} (1.00), i_{dc} (0.80), i_{dr} (0.72)	$-89.0 \pm j99.2$	15.8	66.8	v_{DC} (1.00), i_{dr} (0.80), i_{dc} (0.73)
8	$-106.7 \pm j56.5$	8.9	88.3	i_{dc} (1.00), v_{DC} (0.79), $\varphi_{i_{dc}}$ (0.77)	-141.4	0.0	100.0	β (1.00)
9	$-71.9 \pm j55.4$	8.8	79.2	i_{qr} (1.00), $\varphi_{i_{qr}}$ (0.95)	$-111.1 \pm j58.7$	9.3	88.4	i_{dc} (1.00), v_{DC} (0.78), $\varphi_{i_{dc}}$ (0.73)
10	-77.1	0	100	$\varphi_{i_{qc}}$ (1.00)	$-72.1 \pm j59.6$	9.5	77.1	i_{qr} (1.00), $\varphi_{i_{qr}}$ (0.76)
11	$-29.9 \pm j22.7$	3.6	79.6	θ_{PLL} (1.00), ω_{PLL} (0.99)	-77.1	0.0	100.0	$\varphi_{i_{qc}}$ (1.00)
12	$-44.5 \pm j7.1$	1.1	98.7	$\varphi_{v_{DC}}$ (1.00), v_{DC} (0.83)	$-29.9 \pm j22.6$	3.6	79.7	θ_{PLL} (1.00), ω_{PLL} (0.99)
13	-32.4	0.0	100	φ_{Q_c} (1.00)	$-43.8 \pm j6.2$	1.0	99.0	$\varphi_{v_{DC}}$ (1.00), v_{DC} (0.79)
14	$-4.0 \pm j5.1$	0.8	61.7	θ_s (1.00), $\varphi_{v_{qs}}$ (0.95)	-32.9	0.0	100.0	φ_{Q_c} (1.00)
15	$-1.3 \pm j1.4$	0.2	69.1	φ_{Q_s} (1.00), $\varphi_{v_{ds}}$ (0.76)	-20.0	0.0	100.0	ω_2 (1.00)
16	-0.2	0	100	ω_t (1.00)	$-3.6 \pm j6.6$	1.0	48.2	θ_s (1.00)
17	-0.1	0	100	T_g (1.00)	$-2.4 \pm j1.1$	0.2	91.3	φ_{Q_s} (1.00), i_{qt} (0.74), $\varphi_{v_{qs}}$ (0.72), i_{qg} (0.71)
18	-0.0	0	100	i_{dg} (1.00), i_{qg} (1.00)	-0.2	0.0	100.0	ω_t (1.00)
19					-0.1	0.0	100.0	T_g (1.00)
20					-0.0	0.0	100.0	i_{dg} (1.00), i_{qg} (1.00)

for the GFL-based DFIG, i_{dc} and $\Phi_{i_{dc}}$ (indicated by letter K). Two modes that shift slightly toward the right half-plane are associated with the GSC inner-loop of the GFM-based DFIG, i_{dc} , v_{DC} , and $\varphi_{i_{dc}}$ (marked by letter J) and network current dynamics, i_{qt} , i_{dt} , i_{ds} , and i_{qs} (marked by letter F); however, the reduction is minimal across the entire scenario and does not pose any stability concern. On the other hand, a closer inspection of the highlighted area reveals that certain modes shift further toward the right-half plane compared to the previous case, which indicate a greater risk of instability.

For instance, the modes labeled B and C are associated with the speed ω_s and phase angle θ_s dynamics of the APC in the GFM controller, respectively. This behavior is consistent with the known characteristic of GFM inverters to become less stable as grid strength increases [48], in contrast to GFL inverters which in general exhibit improved stability under stronger grid conditions as discussed in [49].

Furthermore, it is interesting to observe the impact of varying the X_g/R_g ratio on system stability while keeping the short-circuit ratio (SCR) fixed at 10. The X_g/R_g ratio

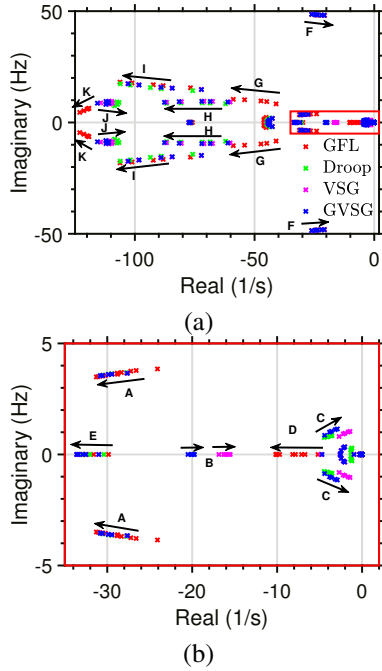


FIGURE 8. Eigenvalue trajectories due to different grid strength: from weak-grid to strong-grid with the same $X_g/R_g = 10$. (a) Complete Plot and (b) Zoomed-in Plot.

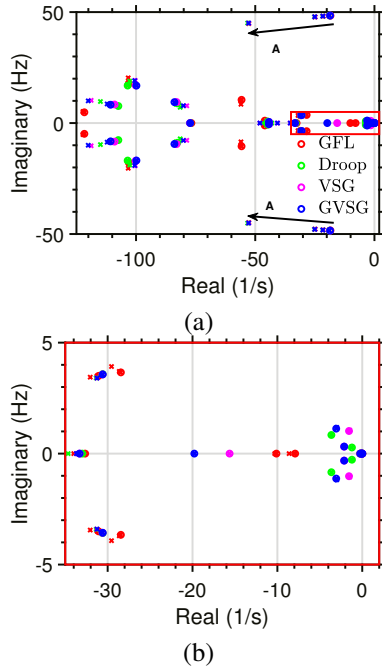


FIGURE 9. Eigenvalue trajectories due to different value of X_g/R_g . The SCR is maintained at 10. (a) Complete Plot and (b) Zoomed-in Plot.

is varied across values of 100, 50, 10, 5, and 1, representing a gradual transition toward a more resistive grid. The eigenvalue trajectories for this scenario are shown in Fig. 9. In general, varying the X_g/R_g ratio has minimal impact on most eigenvalues across all control modes. For example, this can be observed in the APC states of the GFM controller where ω_s and θ_s remain relatively unchanged across all variations

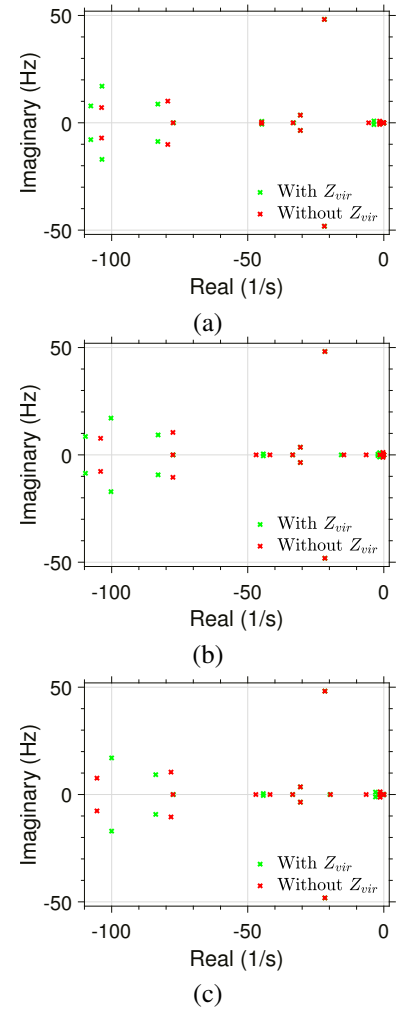


FIGURE 10. Comparison of eigenvalues under different GFM control strategies with and without Z_{vir} . (a) Droop, (b) VSG, and (c) GVSG.

of the X_g/R_g ratio. In contrast, the network current states exhibit a strong dependence on the X_g/R_g ratio as indicated using letter A in Fig. 9. With a decreasing X_g/R_g ratio and hence a less inductive system, the associated current modes exhibit improved stability.

C. IMPACT OF VIRTUAL IMPEDANCE Z_{VIR}

In this scenario, the impact of virtual impedance Z_{vir} on the small-signal stability of a DFIG under various GFM control strategies is investigated. The value of Z_{vir} is set to $0.105 + j0.323$ as given in Table 2, whereas the SCR is set to 10 to emulate a strong grid condition with the grid impedance ratio fixed at $X_g/R_g = 100$. This configuration is considered due to the fact that instability is frequently observed when GFM inverters operate under strong grid conditions.

Fig. 10 presents the results, including a comparison with the eigenvalues obtained without Z_{vir} . In general, the inclusion of Z_{vir} to the control structure enhances the small-signal stability of the DFIG across all GFM control strategies. This improvement is evidenced by the shift of most eigenvalues

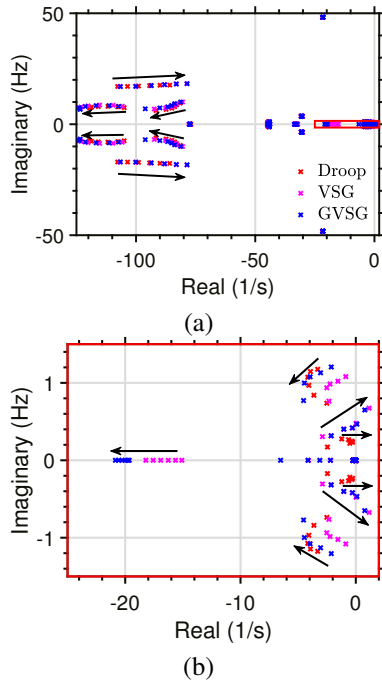


FIGURE 11. Eigenvalue trajectories due to different value of Z_{vir} . The SCR is maintained at 10. (a) Complete Plot and (b) Zoomed-in Plot.

further into the left-half plane when Z_{vir} is applied. In particular, Z_{vir} significantly improves stability of the low-damping modes associated with the APC dynamics, particularly those related to θ_s and ω_s . For instance, under droop control, the relevant modes shift from $-1.53 \pm j4.84$ to $-3.63 \pm j5.28$. In the case of VSG control, the modes move from $-0.27 \pm j7.09$ and -14.69 to $-1.36 \pm j7.98$ and -19.60 , respectively. Similarly, for GVSG control, the modes shift from $-1.36 \pm j7.98$ and -19.59 to $-3.05 \pm j7.10$ and -19.80 , respectively. Thus, GVSG observes the most significant shift in eigenvalues, indicating high sensitivity to the incorporation of Z_{vir} . These shifts indicate a notable improvement in damping, particularly for the APC-related modes. The findings are consistent with those reported in [41], [42]. These observations highlight the potential of virtual impedance as a practical way to improve stability of GFM-controlled DFIGs.

To gain deeper insight into the influence of Z_{vir} on system stability, the value, as specified in Table 2, is varied from 50% to 300% of this nominal setting. The results of the parameter variation are depicted in Fig. 11. As shown, some modes with real parts greater than -80 are significantly affected in an adverse manner by the variation in Z_{vir} , however, this is not considered to pose a threat to stability, as the modes remain well-damped throughout the scenario. Focusing on the zoomed-in region, it is observed that increasing the value of Z_{vir} can lead to instability in certain modes. For instance, at 300% of the base value, two unstable eigenvalues emerge for both VSG and GVSG controls, located at $1.15 \pm j4.23$ and $0.77 \pm j4.08$, respectively. Based on the participation factor analysis, these unstable modes are primarily associated with the RSC controller, including the APC dynamics.

Specifically, they involve states such as θ , $\varphi_{v_{ds}}$, φ_{Q_s} , and $\varphi_{v_{qs}}$, each involving the instability with varying degrees of participation. For droop control, overall stability is maintained, although one mode at $-0.51 \pm j1.37$ approaches the imaginary axis, indicating reduced damping. This mode is primarily associated with the RSC dynamics, particularly the states $\varphi_{v_{ds}}$ and φ_{Q_s} . In contrast, the APC-related mode θ remains well-damped, located at $-3.31 \pm j7.38$, with a damping ratio of approximately 40%, indicating no critical stability concern for this mode under this condition.

D. IMPACT OF INNER-LOOP CURRENT CONTROLLER

This scenario investigates the role of the inner-loop current controller in enhancing the stability of GFM DFIGs. To demonstrate this effect, a comparison is made between the eigenvalue analysis of the DFIG with and without the inner current controller. Specifically, the inner current controller of the GFM RSC as in Fig. 1 is removed, while the outer-loop voltage controller and the virtual impedance are retained. A similar configuration is described in [50]. Furthermore, the control structure for the GSC remains unchanged.

The results of this comparison for all GFM control strategies is provided in Fig 12. Here, only the eigenvalues with the real parts smaller than -50 are shown. The results clearly indicate that the network current modes are modestly influenced by the inner-loop controller, with the modes without the current control exhibiting slightly better damping. A more interesting result can be seen in the RSC-related modes. Generally, these specific modes in the system equipped with the current controller exhibit better damping than those in the system without the current controller. Moreover, the participating states associated with these modes differ between the two systems in all control modes. For instance, in VSG control with a current controller, the critical modes are located at $-1.55 \pm j6.42$ and $-2.14 \pm j1.98$, with the former associated with θ and the latter corresponding to φ_{Q_s} and $\varphi_{v_{qs}}$. In contrast, the case without current control exhibits critical modes at $-1.23 \pm j32.66$, associated with $\varphi_{v_{ds}}$ and $\varphi_{v_{qs}}$, and at $-5.22 \pm j7.45$, associated with θ and ω . Therefore, the absence of the current controller clearly results in a less stable system.

To carry out additional analysis of impact of the RSC's current controller on the system stability, the control parameters namely K_{pir} and K_{iir} are varied from the nominal values given in Table 2. The variation is performed according to the scenarios as follows: first, K_{iir} is set fix at 20, while K_{pir} is varied from 0.5 to 3 times of the nominal value, second, K_{pir} is set fix at 0.3, while K_{iir} is varied from 0.5 to 3 times of the nominal value. The results for these two cases are presented in Fig. 13. In summary, the variations of K_{pir} and K_{iir} affect the network current modes with K_{pir} having a stronger influence, whereas, the RSC-related control modes are only slightly affected. Specifically, the increase on K_{pir} consistently drives the network current modes to the right, reducing damping of the modes. This degrades system stability despite faster rotor's current regulation responses. Subsequently, this

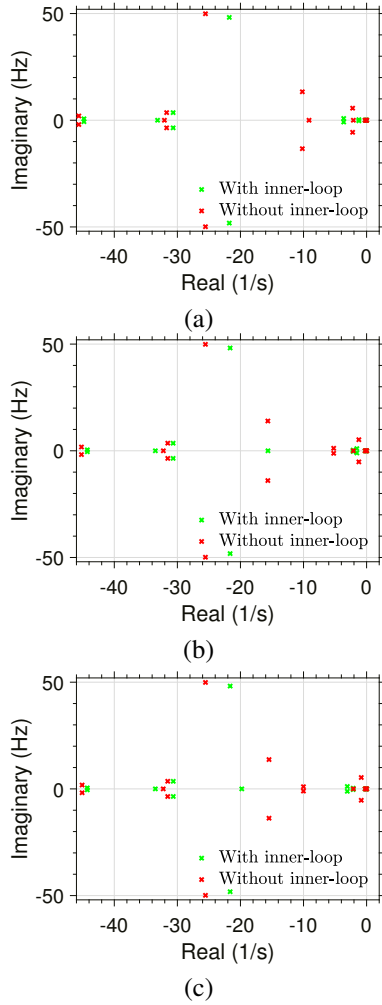


FIGURE 12. Comparison of eigenvalues under different GFM control with and without inner-loop current controller. (a) Droop, (b) VSG, and (c) GVSG.

aggressive current regulation might adversely affect grid dynamics, and thus reduces overall stability. On the other hand, the damping of the network current modes improves as K_{iir} increases. However, these modes are relatively less sensitive to variations in K_{iir} compared to K_{pir} , suggesting a weaker dynamic coupling through the integral action of the current controller. This increase in damping may be attributed to the fact that K_{iir} enhances low-frequency rotor current regulation, enabling the RSC to more effectively suppressing oscillations in the network current.

E. IMPACT OF OUTPUT ACTIVE POWER LEVEL

This section analyzes the impact of DFIG's output active power on system stability. To carry out this analysis, the output active power of the DFIG under all GFM control strategies is varied from 0.1 to 0.95 pu, representing operating conditions from light to heavy loading, while the output reactive power is maintained at zero for unity power factor operation. The SCR is maintained at 10 for a strong grid simulation.

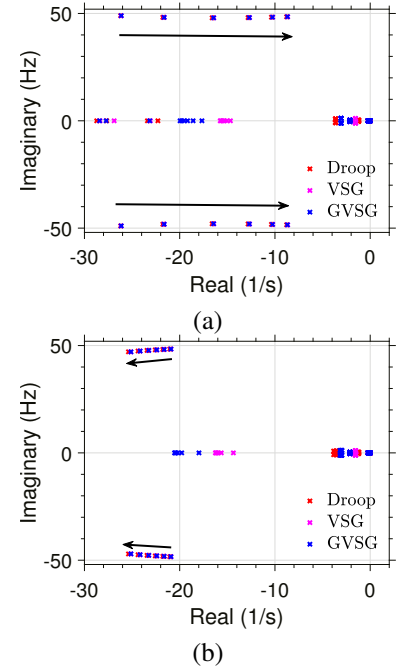


FIGURE 13. Eigenvalue trajectories for varying values of the RSC's current controller parameters: (a) K_{pir} and (b) K_{iir} .

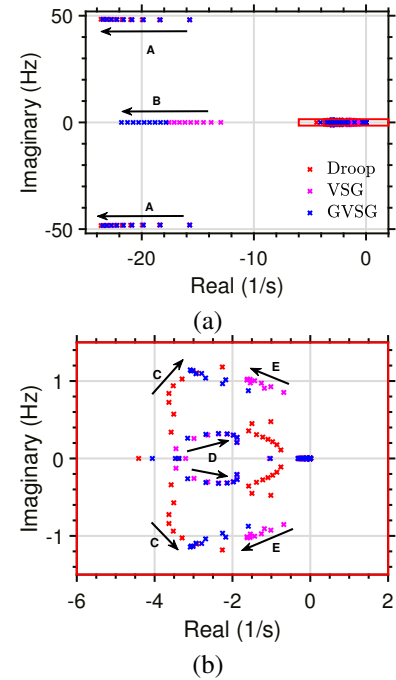


FIGURE 14. Eigenvalue trajectories due to different value of the DFIG's output active power from 0.1 to 0.95 pu at unity power factor operation. (a) Complete Plot and (b) Zoomed-in Plot.

The results are illustrated in Fig. 14. In summary, increasing the output active power influences the eigenvalues of the GFM DFIG at varying rates, depending on the mode. Notably, the output active power level exerts a consistently similar effect on the network current modes, denoted by letter A, across all control strategies. As the output active power

TABLE 4. Unstable Eigenvalues and Participation Factor during Off-grid scenario

Control	Eigenvalues λ_i	f_i	ζ_i	States & p_{ki}
Droop	3.2×10^{-3}	0.00	-100	$i_{dr}(1.00), i_{ds}(0.89), \omega_t(0.86), i_{qL}(0.70)$
VSG	3.1×10^{-3}	0.00	-100	$i_{dr}(1.00), \omega_t(0.98), i_{ds}(0.89), i_{qL}(0.70)$
GVSG	3.2×10^{-3}	0.00	-100	$i_{dr}(1.00), \omega_t(0.98), i_{ds}(0.89), i_{qL}(0.70)$

increases, these modes exhibit improved damping. This phenomenon can be explained as follows: with increasing real power output, the rotor speed ω_r rises, leading to a decrease in slip, defined as $s = \omega_s - \omega_r$. Consequently, the rotor resistance referred to the stator, $R'_r = R_r/s$, increases, thereby raising the total effective impedance of the system. This increase in impedance enhances the damping of the network current modes, as shown in the plot.

Focusing on the zoomed-in plot, the effect of varying the DFIG's real power output shows a similar trend for the VSG and GVSG strategies, while a different behavior is observed for Droop. In the Droop control case, as the output active power increases, two modes indicated by letters C and F exhibit reduced damping, shifting from $-3.63 \pm j5.28$ and $-1.22 \pm j1.73$ to $-1.56 \pm j7.66$ and -1.11 , respectively. These modes are related to the RSC control dynamics, including θ , φ_{vds} , φ_{vqs} , and φ_{Qs} . Despite the reduction in damping, the system retains stability under all loading conditions. A more complex pattern emerges in the VSG and GVSG strategies, as indicated by letters D and E in Fig. 14. The former, associated with φ_{Qs} and φ_{vds} , shows decreased damping with increasing active power output. However, this reduction does not appear to threaten system stability across the considered operating range. In contrast, the latter, related to θ and φ_{vqs} , consistently exhibits improved damping under all loading conditions.

F. OFF-GRID OPERATION

This scenario investigates the eigenvalue behavior of the GFM DFIG under isolated operating conditions. To simulate this condition, the switch S_w in Fig. 1 is opened, effectively resulting in disconnection of the DFIG from the grid. It is to be noted that the APC regulates P_s according to the MPPT algorithm, whereas as the RPC controls Q_s . Further, the grid is modeled as a constant impedance load where power consumed by the load is determined by the load voltage v_{pcc} as given in (32). It is obvious beforehand that this configuration leads to an unstable isolated system as the control structure is not designed to regulate the system's voltage and frequency during active and reactive power unbalance.

To simulate this, the load is set at $P_L = 0.3$ pu and $Q_L = 0.3$ pu. From the load flow analysis, it is found that $v_s = 1.02 \angle 0^\circ$ and $v_{pcc} = 0.93 \angle -4.1^\circ$, and hence, the load impedance can be computed as follows: $Z_L = R_L + jX_L = |v_{pcc}|^2 / (P_L - jQ_L) = 1.43 + j1.43$ pu. The

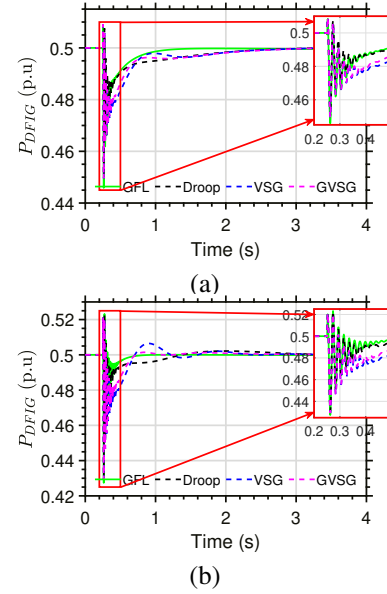


FIGURE 15. Active power output responses of DFIG due to a voltage increase at the infinite bus. (a) weak grid and (b) strong grid.

unstable eigenvalues for all control strategies along with the participating states are given in Table 4.

Several observations can be made regarding these unstable modes. First, the modes exhibit purely real eigenvalues, indicating that the instability is non-oscillatory in nature. Second, across all control strategies, the dominant participating states in the unstable modes are consistently i_{dr} , i_{ds} , ω_t , and i_{qL} . These variables are primarily associated with the MPPT control loops. This behavior can be explained as follows. Under off-grid conditions, the load S_L is supplied solely by the DFIG. During a grid disturbance, the power consumed by the load is determined by the load voltage, leading to an active power imbalance. However, the APC continues to regulate the DFIG's output based on the MPPT algorithm as in (6), without adapting to the load consumption. If this mismatch persists, the system becomes unstable. One way to solve this problem is to augment the control structure with frequency and voltage regulation mechanisms as proposed in [51].

VIII. TIME-DOMAIN SIMULATION

This section presents a detailed time-domain analysis to further assess the stability of GFM DFIG systems under various operating conditions. The objective is to complement the small-signal analysis in the previous section by capturing nonlinear dynamic behavior responses.

A. VOLTAGE DROP AT THE INFINITE BUS

This scenario simulates the nonlinear transient responses of DFIG systems following a small-signal disturbance. For this purpose, the DFIG system shown in Fig. 1 is subjected to a 5% grid voltage drop at $t = 0.25$ s under all four control strategies, namely GFL, Droop, VSG, and GVSG. The simulation is carried out for 4 s and considers two grid

strength conditions, namely a weak grid with SCR = 2 and a strong grid with SCR = 10. The responses of DFIG's active power output during weak and strong grid conditions are provided in Figs.15a and 15b, respectively.

The results show a pronounced oscillatory behavior in the aftermath of the disturbance, but the oscillations are damped quickly within approximately 0.5 s, characterizing a high damping factor for the modes. Investigating the time-domain response in the zoomed-in plot reveals that the oscillation frequency is approximately 50 Hz, which clearly corresponds to the network current modes previously analyzed in Section VII-A. The waveform responses across all control strategies appear quite similar, indicating the fact that the modes are not significantly affected by the choice of control strategy. Moreover, comparing the time responses between the weak and strong grid conditions suggests that the network current modes are also not significantly influenced by grid strength, consistent with the finding in Section VII-B.

A further investigation on Fig.15 demonstrates that there exist low-frequency oscillations in particular in VSG and GVSG control schemes with the frequency around 1 Hz. The analysis previously carried out in Section VII-A indicates that these modes are the low-damping modes related to the APC state θ (see λ_{14} in the VSG case and λ_{16} in GVSG in Table 3). The damping of these modes degrades, indicated by more oscillatory behavior, as the grid strength increases, which corresponds to the well-known behavior of GFM inverters that tend to perform poorly under strong grid conditions [48]. This observation is consistent with the analysis in Section VII-B.

B. IMPACT OF VIRTUAL IMPEDANCE ON TIME-DOMAIN RESPONSES

This scenario investigates the influence of the virtual impedance on the time-domain responses of GFM controlled DFIGs. The analysis is performed by comparing the waveforms of the RSC controller with and without the inclusion of Z_{vir} across all control strategies in response to disturbances. The simulation setup is as follows. The system's SCR is set to 10 to represent a strong grid condition, where virtual impedance is typically employed to increase the equivalent impedance. The system is again perturbed by a 5% voltage drop at the infinite bus for a 2.5 s simulation. The active power output for all GFM control configurations is depicted in Fig. 16.

Investigating closely the detailed portion of the plot clearly indicates that the network current modes with frequency around 50 Hz are barely affected by the absence of Z_{vir} in virtually all control scenarios. This is indicated by the close resemblance of the waveforms immediately following the fault. The observation is in line with the previous study in Section VII-C, where the eigenvalues corresponding to the network current modes show minimal variation for both the cases with and without Z_{vir} .

In contrast, the APC-related modes with frequencies around 1 Hz are significantly influenced by the inclusion of

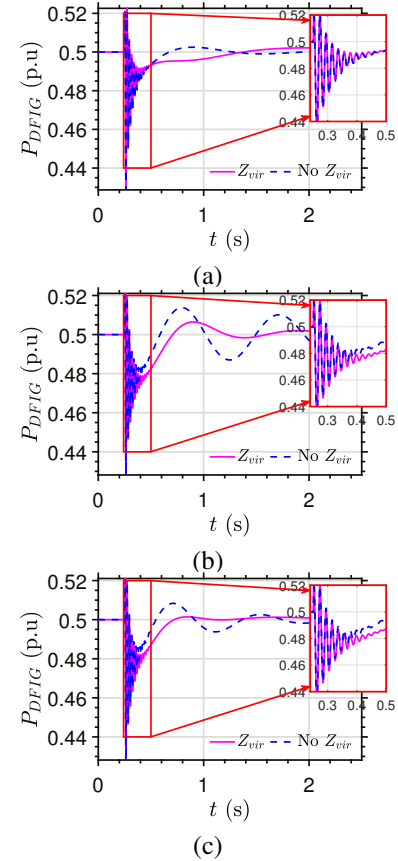


FIGURE 16. Active power output responses of GFM DFIG system with and without virtual impedance Z_{vir} . (a) Droop, (b) VSG, and (c) GVSG.

Z_{vir} . The addition of virtual impedance leads to a substantial improvement in the damping of these low-frequency oscillations. Among all control strategies, GVSG exhibits the most substantial increase in damping, characterized by a significant difference in the waveforms for both the cases, whereas the droop control shows the least improvement in the time-domain response (see Fig. 16). This consistent observation also aligns with the findings in Section VII-C, where the APC-related eigenvalue in the GVSG case shifts noticeably toward the left-half plane after incorporating Z_{vir} into the control loop, while the corresponding eigenvalue in the droop controller shows only a minimal shift.

C. THE IMPACT OF INNER-LOOP CURRENT CONTROLLER ON TIME-DOMAIN RESPONSES

This scenario offers a complementary analysis to the study in Section VII-D, specifically focusing on the influence of the RSC's current controllers on transient behavior of the DFIG with different GFM control schemes. Modal analysis previously revealed that the absence of the RSC current controllers degrades system damping, particularly in the APC-related modes. This section, thus, aims to demonstrate how this phenomenon manifests in the time-domain responses.

With the aim to examine the impact of the RSC's current control in the time-domain, the GFM DFIG systems both

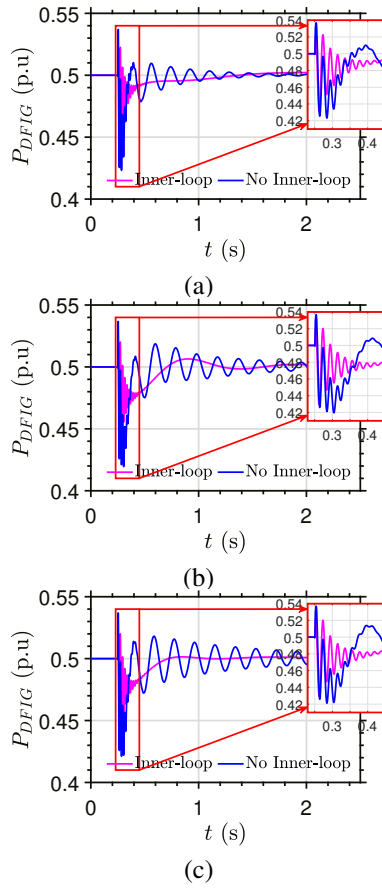


FIGURE 17. Active power output responses of GFM DFIG system with and without the current loops. (a) Droop, (b) VSG, and (c) GVSG.

with and without the RSC current controller are subjected to a 5% voltage drop at the infinite bus at $t = 0.25$ s. The virtual impedance Z_{vir} and the outer-voltages are retained for all systems. The waveform of DFIG's real power output in response to the fault is given in Fig. 17, which will be analyzed in terms of oscillatory behavior and settling-time. On the whole, the absence of the current controllers results in more pronounced oscillations in the time-domain responses compared to the case with the controller. In particular, the activity of the network current modes, immediately after the appearance of the fault, is similar in both cases; however, the case without the inner loops exhibits slightly better damping and is completely damped out before $t = 0.35$ s. For the APC-related modes, investigating the sustained oscillations clearly shows the frequency at around 5 Hz, with the oscillations persisting until the end of simulation. From the modal analysis in Section VII-D, this mode is associated with $\varphi_{v_{ds}}$ and $\varphi_{v_{qs}}$. The relatively low damping of this mode definitely contributes to the persistent oscillations observed in the time-domain response. The stability of GVSG is strongly influenced by the current controllers, whereas the droop control shows minimal sensitivity to their presence. This finding signifies the critical role of the inner-loop current controller of GFM DFIG control under disturbances. Improving the

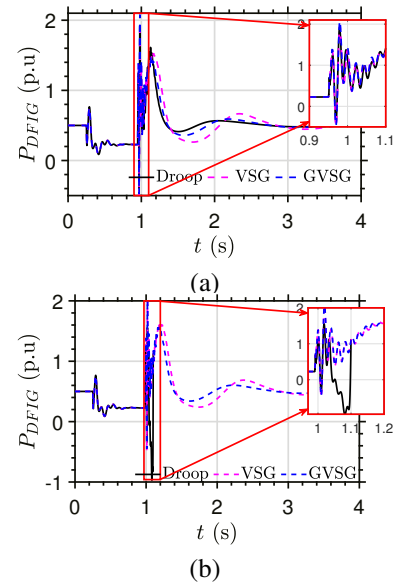


FIGURE 18. Active power output responses of GFM DFIG system during off-grid conditions for different re-synchronization times. (a) $t = 0.95$ s and (b) $t = 1.00$ s.

damping of this mode via tuning of PI control parameters should significantly enhance transient performance.

D. PERFORMANCE DURING OFF-GRID CONDITION

A key limitation of GFL DFIGs is their inability to operate under islanded conditions, as their control architecture including both RSC and GSC mainly depends on a PLL to track the voltage angle reference. In off-grid scenarios, the absence of a voltage reference prohibits the effective operation of the PLL. This limitation can be addressed by GFM inverters, which mimic synchronous generator's behavior by synthesizing an internal voltage angle independently. This study aims to perform an investigation on the off-grid capability of GFM DFIGs under different control schemes.

To simulate this scenario, the test system in Fig. 1 is initially connected to the grid, delivering power $S_{DFIG} = 0.5 + j0$ pu to supply a load of $S_L = 0.2 + j0.1$ pu. The surplus power is exported to the grid. At $t = 0.25$ s, the DFIG is disconnected from the grid by opening the switch S_w , and subsequently re-synchronized. Two synchronization scenarios are considered, namely Scenario-1 at $t = 0.95$ s and Scenario-2 at $t = 1.00$ s. The results are shown in Fig. 18.

In Scenario-1, it is clear that the DFIG is successfully re-synchronized to the grid under all three GFM control algorithms, with the VSG exhibiting slightly more oscillatory behavior, whereas the droop control shows the least oscillatory response. Despite this, the oscillations in all control algorithms subside within 3 s, indicating that each GFM strategy maintains acceptable transient stability following re-synchronization. Note that GFL DFIG is not considered in this scenario due to the fact that the system is inherently unstable. In Scenario-2, where the re-synchronization

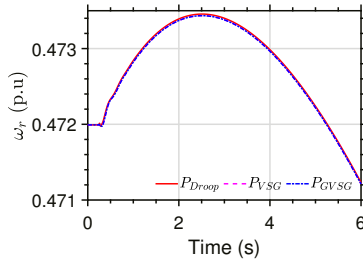


FIGURE 19. Rotor's speed ω_r responses of GFM DFIG during islanded operations due to a load change.

is performed at $t = 1.00$ s, both the VSG and GVSG controllers successfully restore grid connection and maintain stable operation. In contrast, the droop-controlled DFIG fails to achieve stable synchronization and exhibits instability following reconnection. This demonstrates a limitation in the droop control strategy, which may fail to ensure stable re-synchronization under certain timing conditions. Such operation constrain should be carefully considered in the context of wind farm operation, particularly when grid reconnection events are expected.

Furthermore, it is interesting to see the connection between the findings in Section VII-F and the observed time-domain responses. As previously shown, the fully off-grid DFIG system exhibits instability across all GFM control strategies, with similar non-oscillatory unstable modes identified in Table 4. To see how this manifests in time-domain simulation, the system is perturbed with a slight load increase from $S_L = 0.1 + j0.2$ pu to $S_L = 0.15 + j0.25$ at $t = 0.25$ s. The responses of rotor speed ω_r following this disturbance is given in Fig 19. It is clear that the system is non-oscillatory unstable, consistent with the frequency-domain analysis in Section VII-F. As the load increases, the DFIG remains in the MPPT control strategy and maintains its power output according to (6). This circumstance results in a power shortage condition in the system, leading to the rotor speed droop and subsequent frequency collapse.

E. PERFORMANCE DUE TO INPUT CHANGES

This section investigates the performance of DFIG with different control scenarios in response to input changes. The grid condition is considered strong with SCR=10. Two key scenarios are considered, namely changes in wind speed and stator's reactive power reference Q_s^* . In the first scenario, the wind speed experiences a sudden increase from 11.94 m/s to 12.94 m/s at $t = 1.00$ s. In the second scenario, a step change is applied to the stator reactive power reference Q_s^* , increasing from 0 to 0.1 pu at $t = 0.25$ s. Figs.20a and 20b show the system responses for the first and second scenarios, respectively.

As expected, the change in wind speed leads to an increase in the DFIG's real power output from 0.5 pu to approximately 0.63 pu as given in Fig.20a. The responses are virtually identical across all control modes with the system reaching a new steady-state condition after $t = 15$ s. This suggests

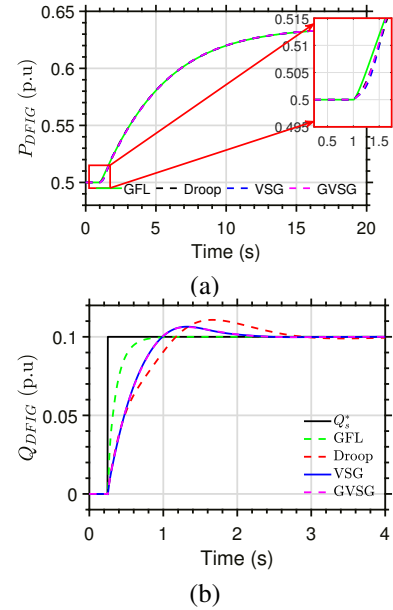


FIGURE 20. Active power output responses of DFIG system due to different input changes. (a) Wind speed change and (b) Q_s^* change.

that the different control strategies have minimal impact on the system's active power tracking under wind speed variations, forcing the DFIG to operate at the maximum power point. Further, the responses to the change in Q_s^* are more interesting as provided in Fig.20b. The GFL exhibits critically damped response, characterized by a monotonic rise toward the reactive power setpoint Q_s^* and reaching the steady-state at $t = 1.00$ s. This might stem from its fast-acting current controllers and well-regulated PLL synchronization under strong grid conditions. On the other hand, all GFM algorithms exhibit underdamped responses characterized by oscillatory behavior and a small overshoot of less than 0.12 pu. This can be caused by their inherent dynamic characteristics, which are designed to emulate synchronous generator behaviors. The VSG and GVSG controllers produce nearly identical responses, both reaching steady-state conditions at approximately $t = 2$ s. In contrast, the droop controller settles slightly later at around $t = 2.5$ s.

F. PHASE JUMP

In the future power systems with high penetration of inverter-based resources, the renewable energy including wind turbines needs to provide grid frequency supports. This means that inverters need to adjust their real power output to maintain grid frequency in a very similar fashion to conventional synchronous generators. This section aims to study the capability of DFIG under different control strategies to provide the frequency support by adjusting the real power outputs. To simulate frequency events in the grid, a sudden increase in grid voltage angle occurs at $t = 0.25$ s. Initially, the grid voltage is set to $v_g = 1.00 \angle 0.0^\circ$ pu and then changed to $v_g = 1.00 \angle 30.0^\circ$.

Fig.21 depicts system responses due to the disturbances.

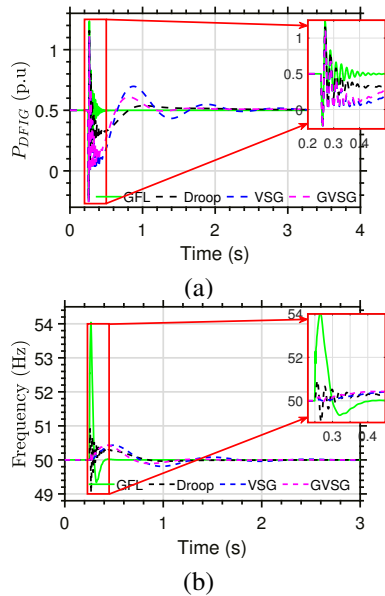


FIGURE 21. Time-domain responses of GFM DFIG due to grid angle voltage jump. (a) P_{DFIG} and (b) Frequency.

In Fig.21a, the GFL DFIG experience a short excursion in P_{DFIG} lasting less than 0.2 s due to fast-acting of current controllers and PLL synchronization. In contrast, the DFIG under GFM controls demonstrates large transients in real power outputs P_{DFIG} to support grid's frequency, temporarily adjusting P_{DFIG} to reduce frequency excursions. Among the GFM control strategies, the droop controller exhibits the least oscillatory response and returns to steady-state at approximately $t = 1.5$ s. The GVSG, while showing larger oscillations than droop, also settles around the same time, indicating improved grid support due to its added virtual inertia and damping as given in (29). In contrast, the VSG experiences a comparable level of overshoot to GVSG, demonstrating a similar grid support, but displays a slower settling response. With this, as given in Fig.21a, the frequency excursions in GFL is the largest reaching 54 Hz with almost no support to the grid. In contrast, the GFM-based strategies effectively limit the frequency deviation and restore the nominal frequency with varying settling times. Among them, the VSG and GVSG controllers exhibit the smallest deviations, with peak values reaching merely 50.04 Hz.

IX. CONCLUSION

This paper has presented a detailed small-signal stability study of three popular GFM control strategies, namely droop, VSG, and GVSG, applied to DFIGs. The comparative study using modal analysis offers new insights into the dynamic behavior of GFM-based wind turbines. It is observed that all GFM control schemes introduces low-frequency modes related to APC dynamics at the frequency around 1 Hz. Furthermore, the performances of all GFM DFIG control modes deteriorates to varying degrees by the increase in grid strength, and the absence of virtual impedance and inner-loop current controller. The stability under both grid-connected

and islanded conditions has also been investigated. Among the three control strategies, in terms of frequency and damping of oscillatory modes, GVSG in general demonstrates superior small-signal stability performance compared to droop and VSG across all scenarios. Time-domain simulations are also carried out to support the findings of frequency-domain analysis.

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